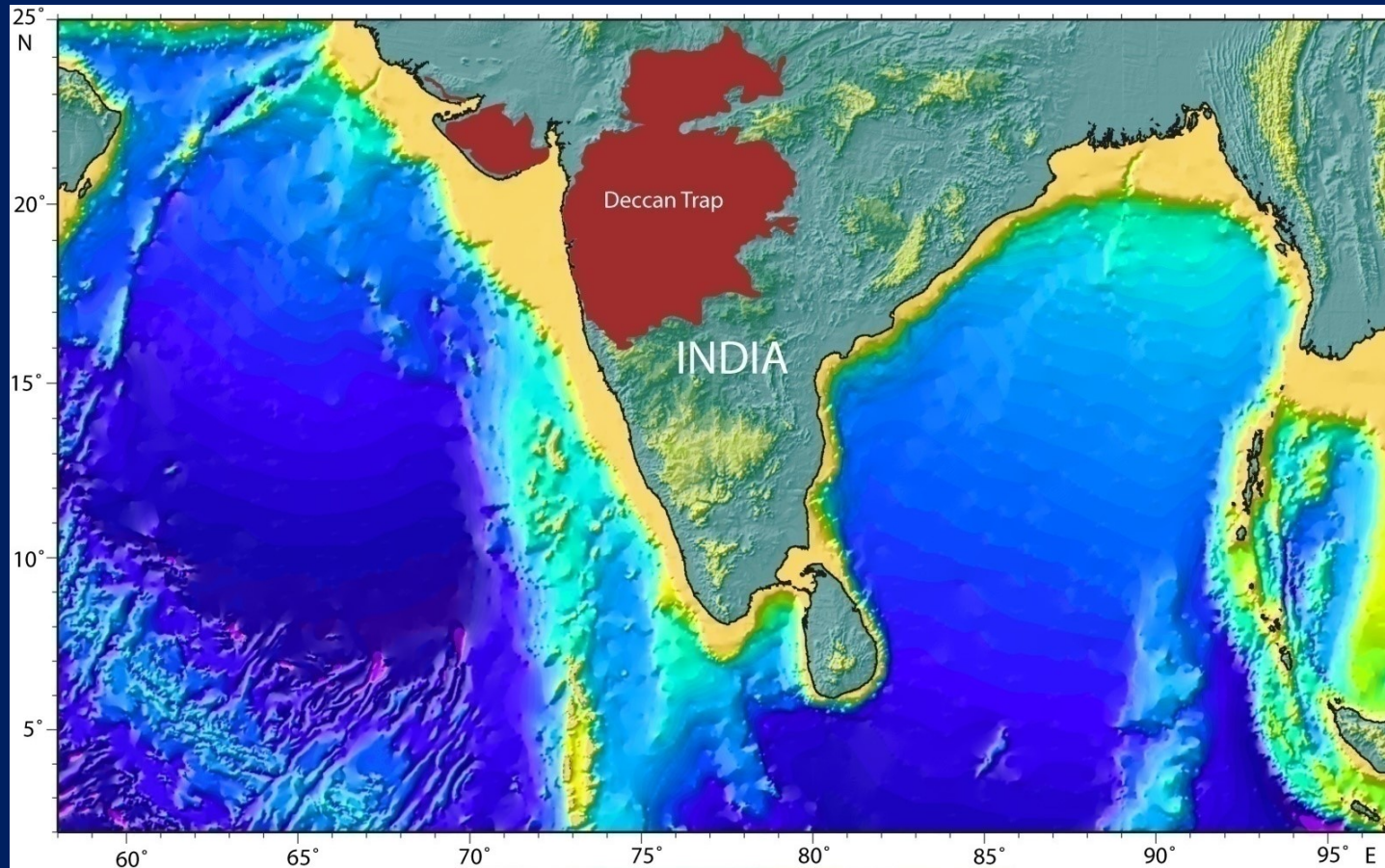


Wind Driven & Thermohaline Circulation



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L35: 10 April 2024

L36: 12 April 2024

Wind Driven Circulation
Thermohaline Circulation
Upwelling & Downwelling

Ocean Circulation: refers to the overall movement of water throughout the oceans. It includes both surface currents and deep-water currents.

Ocean Currents: Ocean currents are more specific, referring to the continuous, directed flow of ocean water in a prevailing direction. They are like rivers within the ocean, flowing at various depths and speeds.

Ocean circulation is the broader term that encompasses all movements of ocean water, while ocean currents specifically refer to the flow of ocean water in a particular direction.

Why is Ocean Circulation Important?

- Transport ~20% of latitudinal heat from equator to poles
- Transport nutrients and organisms matter
- Regulate weather and climate
- Influences commerce
- Carbon Storage
- Regulate Sea-level
- Marine Diversity

**The waters of the ocean is in a
state of continuous motion.**

**What causes oceanic water to
move continuously ?**

Causes of oceanic water to move continuously

- 1. Currents**
- 2. Waves**
- 3. Tides**
- 4. Ocean Swells**
- 5. Littoral Currents**
- 6. Tsunami**

- 1. Currents**
- 2. Waves**
- 3. Tides**
- 4. Ocean Swells**
- 5. Littoral Currents**
- 6. Tsunami**

What are the Forcings?

Waves → Wind

Tides → Gravitational Pull of Moon & Sun

Currents → 1. Winds 2. Heating & Cooling

Ocean Swells → Wind raging over ocean's surface, Gravitational force

Littoral Currents → Coastal Barrier

Tsunami → Land Slide, Earthquakes

Waves

Wind

Tides

**Gravitational Pull of Moon &
Sun**

Currents

1. Winds 2. Heating & Cooling

**Thus, the Ocean Currents and its
circulation can be divided into two parts**

1.Wind-driven Circulation

2.Thermo-haline Circulation

Both of which will be influenced by the

- Rotation of the earth
- Shape of the ocean Basin

Ocean Currents

1. Wind driven → Surface Currents (down to ~ 1km)
Moves water horizontally.
Only 10% of ocean currents moved this way!

2. Density (gravity) Driven → Moves water vertically, is responsible for mixing water masses, and creates thermohaline circulation.

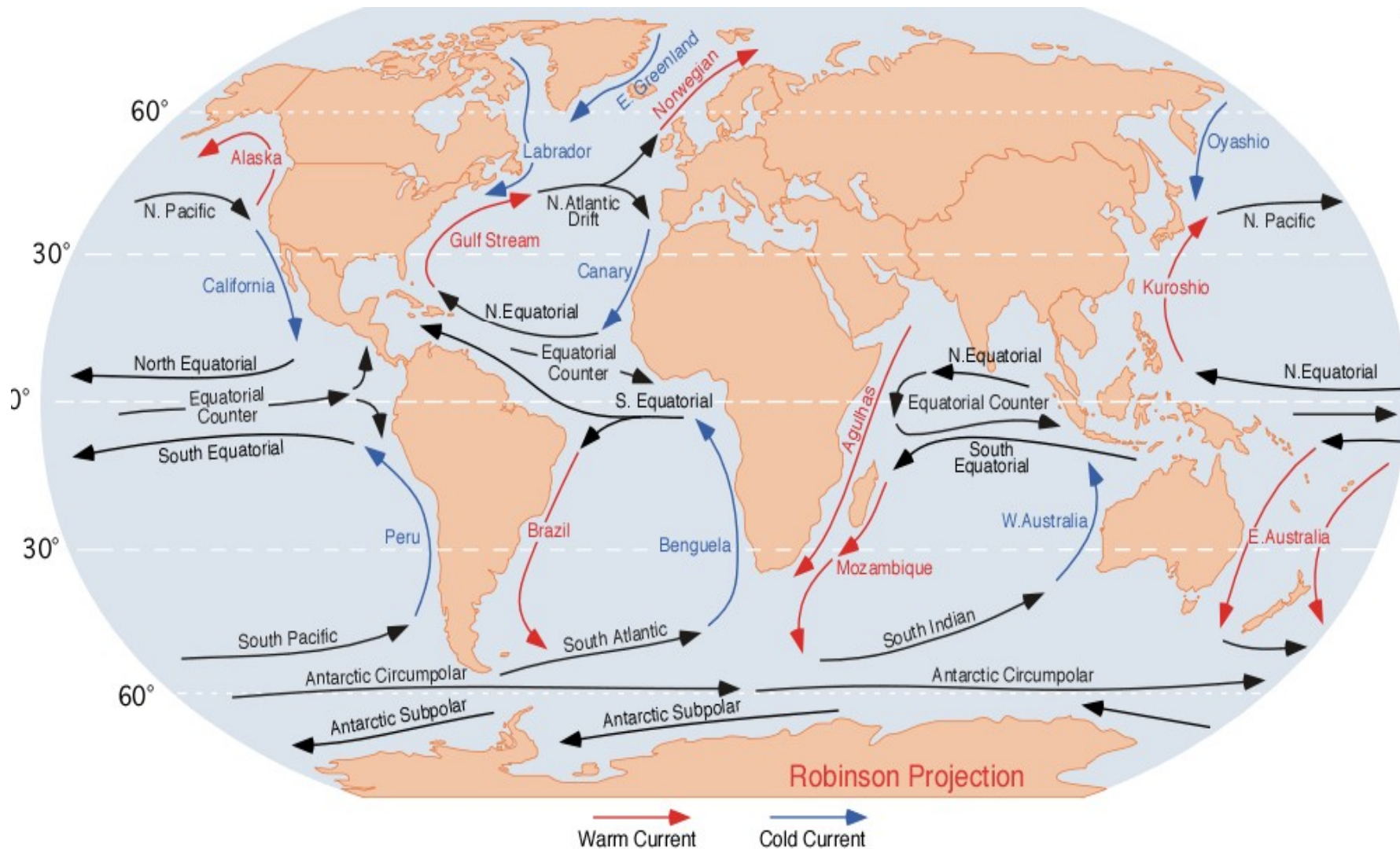
About 90% of ocean currents moved this way!

Surface Currents

Warm Currents: Warm currents are currents that have higher temperatures compared to the surrounding water. They often originate in low latitudes (closer to the equator) where water temperatures are higher and flow towards higher latitudes (closer to the poles). Warm currents can have significant effects on the climate of coastal regions they pass by, such as bringing warmer temperatures and influencing precipitation patterns.

Cold Currents: Cold currents are currents that have lower temperatures compared to the surrounding water. They often originate in high latitudes (closer to the poles) where water temperatures are lower and flow towards lower latitudes (closer to the equator). Cold currents can also have significant climate effects, bringing cooler temperatures and influencing marine ecosystems.

Surface Currents → Warm and Cold Currents



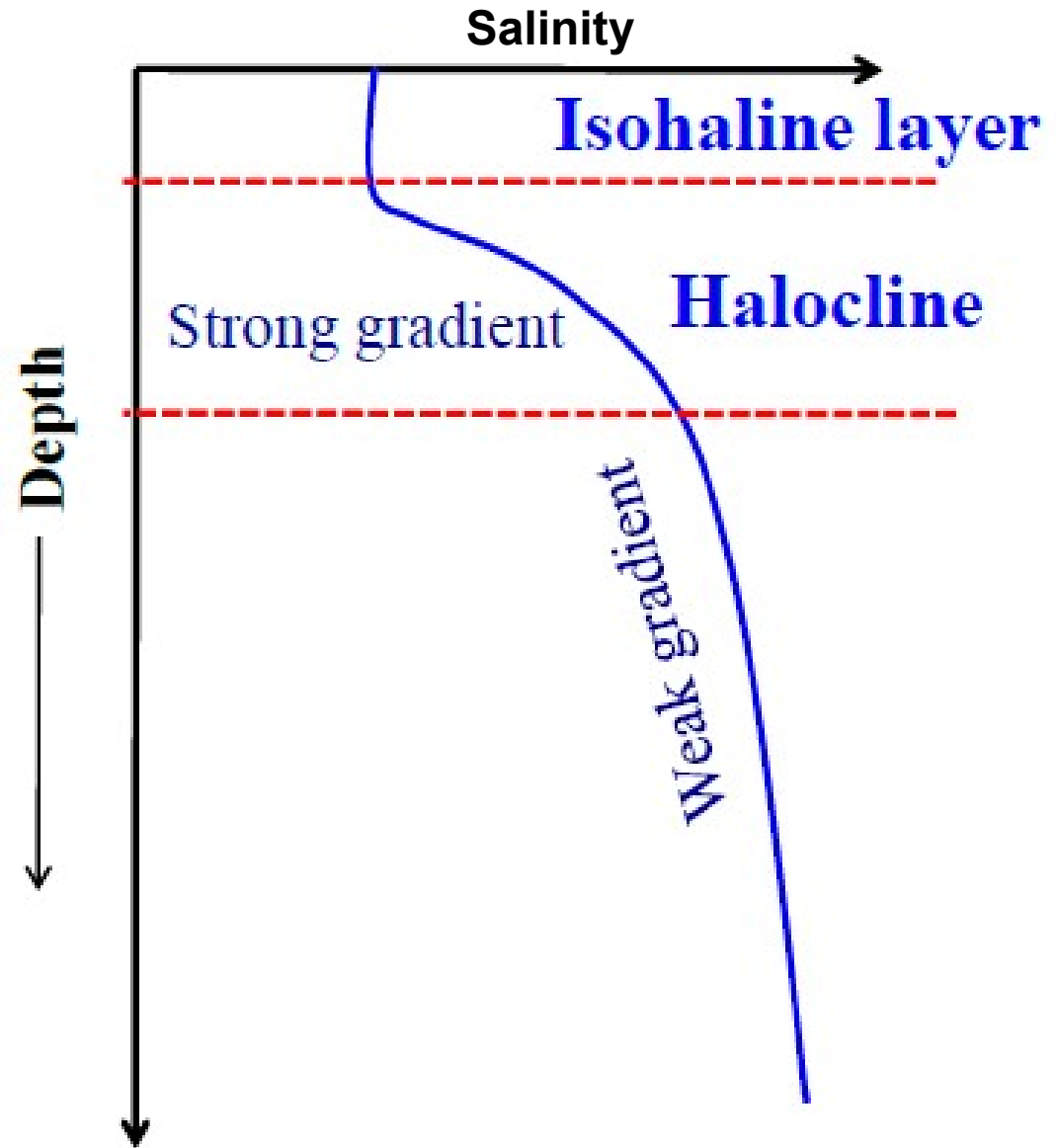
Thermohaline Circulation

- As opposed to wind-driven currents and tides (which are due to the gravity of moon and sun), the thermohaline circulation is that part of the ocean circulation which is driven by density differences.
- Thermohaline circulation (THC) is a part of the large-scale ocean circulation that is driven by global density gradients created by surface heat and freshwater fluxes.
- The word *thermohaline* derived from *thermo-* referring to temperature and *-haline* referring to salt content, factors which together determine the density of sea water.

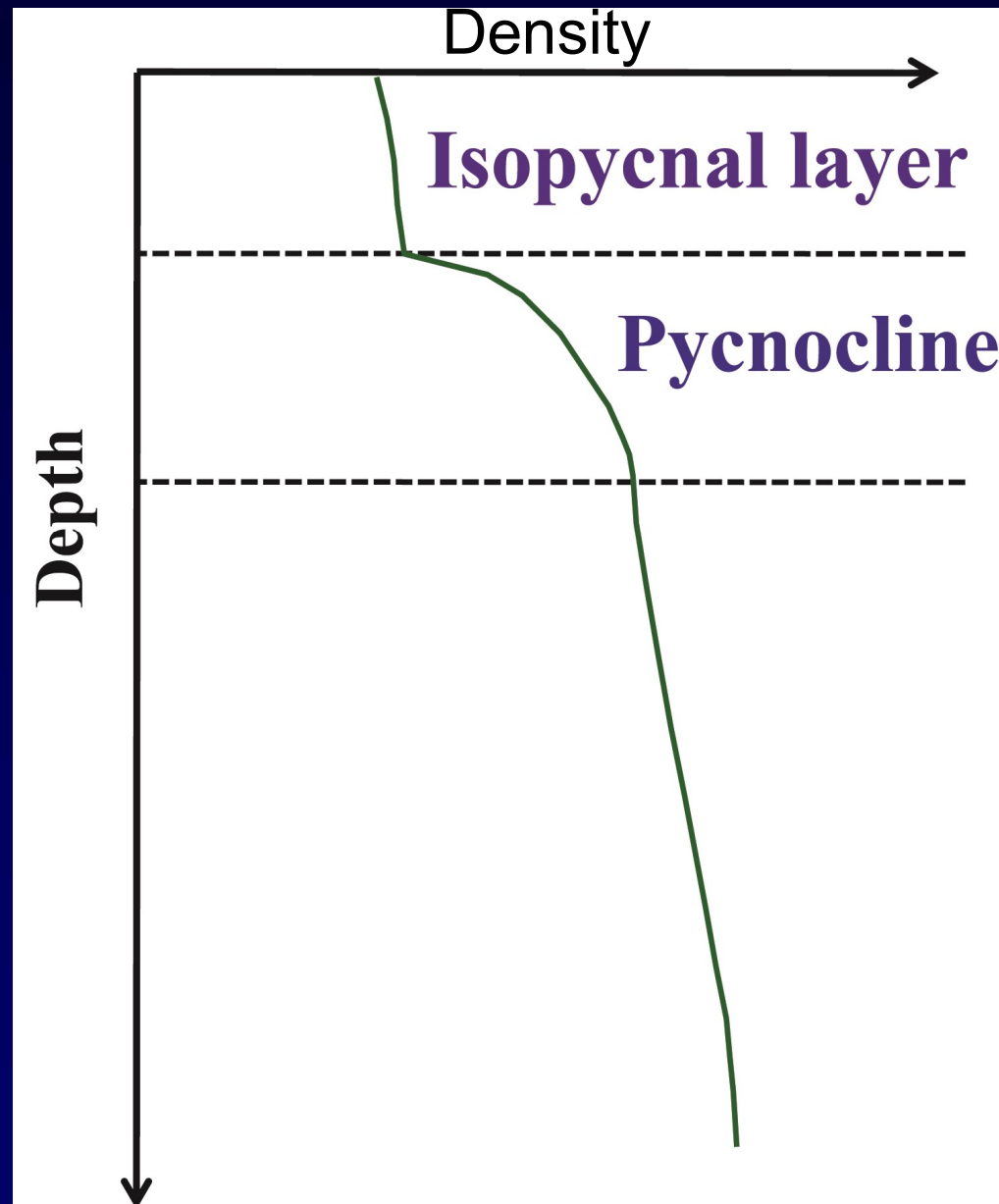
VERTICAL DISTRIBUTION OF SALINITY

Halocline:

It is a vertical zone in the oceanic water column in which salinity changes rapidly with depth, located below the well mixed uniformly saline surface water layer. The greatest change in salinity, with Depth, occurs in Halocline

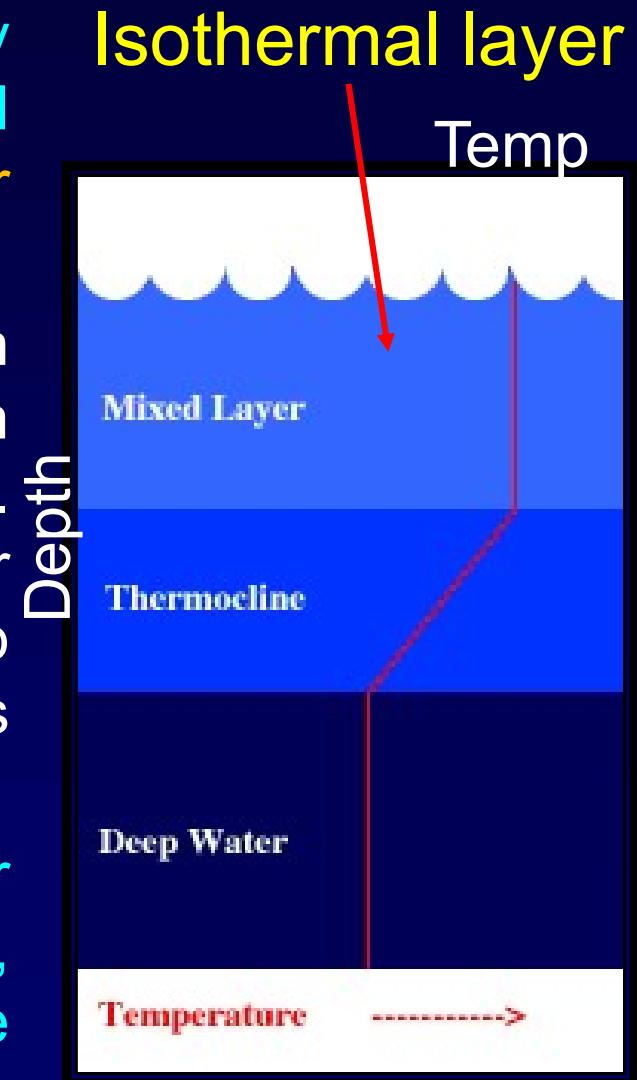


Vertical Profile of Density



Thermal Layers of the Ocean Water

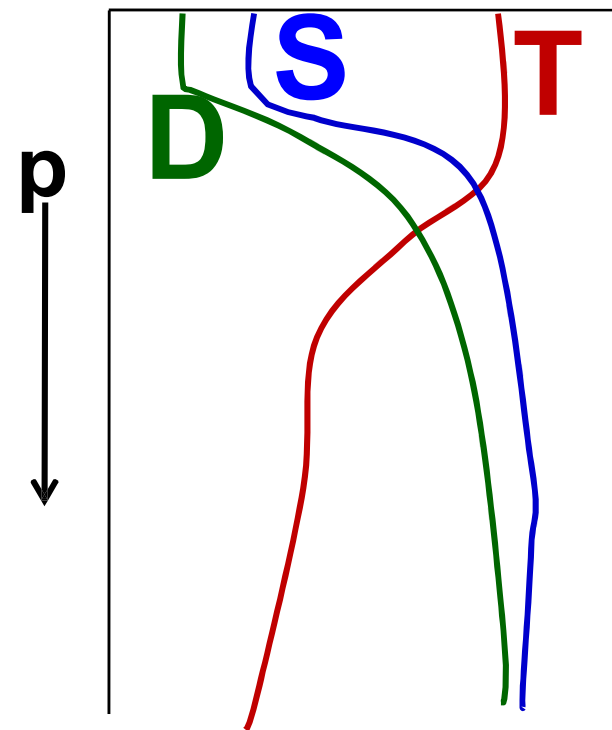
- The turbulent mixing establishes a **isothermal surface-layer** that can vary from 15 to 200 m. Depth at which Mixed layer end is called MLD (**Mixed Layer Depth**)
- Between about 200 m and 1000 m depth temperature declines rapidly. This region of steep gradient is called **Thermocline**. The thermocline is the transition layer between the mixed layer and the deep water layer. Temperature decreases sharply in this layer
- The isothermal layer and the deep water layer are relatively uniform in temperature, while the thermocline represents the transition zone between the two.



Thermohaline Circulation

It refers to the flow of ocean waters caused by changes in sea water **density**.

Since the **Density** of sea water is controlled by its temperature (**Thermo**) and salinity (**haline**), the circulation driven by density differences is called the thermohaline circulation



Processes Controlling Density

Sea water Density could be altered when

1. The ocean is **warmed** or **cooled** at the surface by radiation and contact with atmosphere
2. By addition of **fresh water** (precipitation or river run off or melting of ice)
3. By addition of **salt** (evaporation or ice formation).

Thus, warming & melting of ice would decrease the density while cooling and ice formation will increase the density

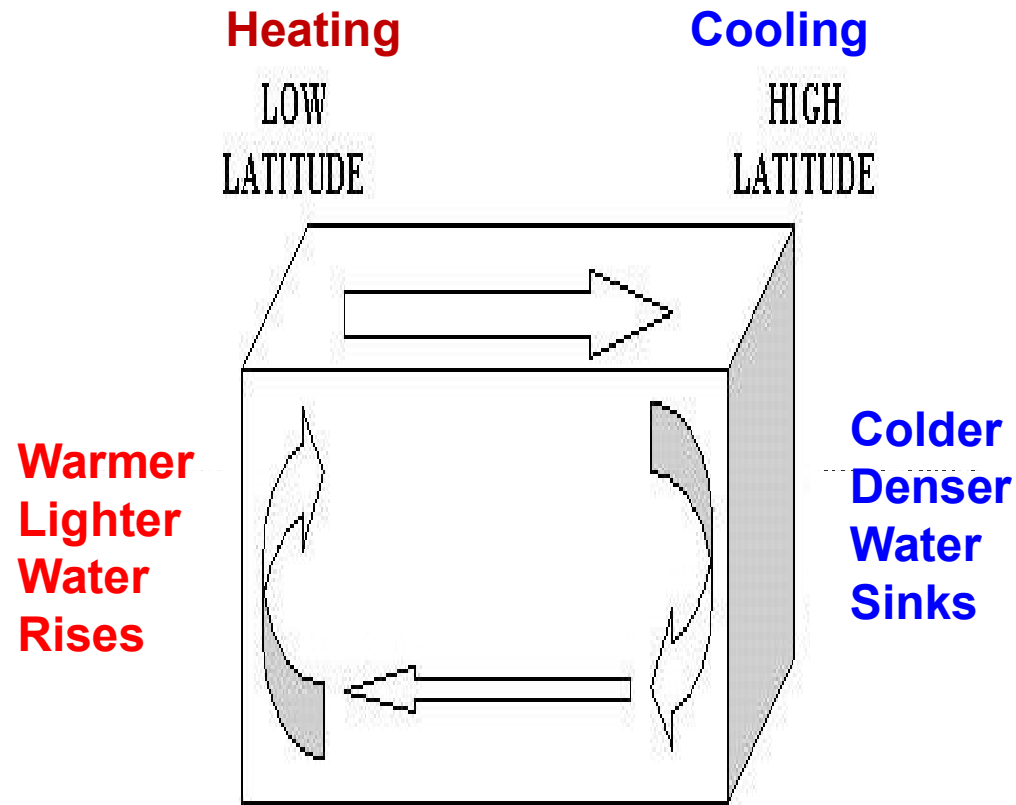
Since the density of sea water depend on **temperature** and **salinity**, **Heating** near the **equatorial region** will decrease the density and **cooling** near **polar region** will increase the density of sea water.

In a **non-rotating earth** the cold and denser Polar water will sink deeper down, while the warm and lighter tropical water will become buoyant and starts flowing from equator towards poles.

Eventually, this will set-up a circulation where in the **cold and denser water** at the polar region will sink and move towards the equatorial region at the deeper layer, while the **warm and lighter water** from the equatorial region will flow towards the polar region along the surface.



This is a simple schematic depicting the thermohaline circulation in a **non-rotating Earth**



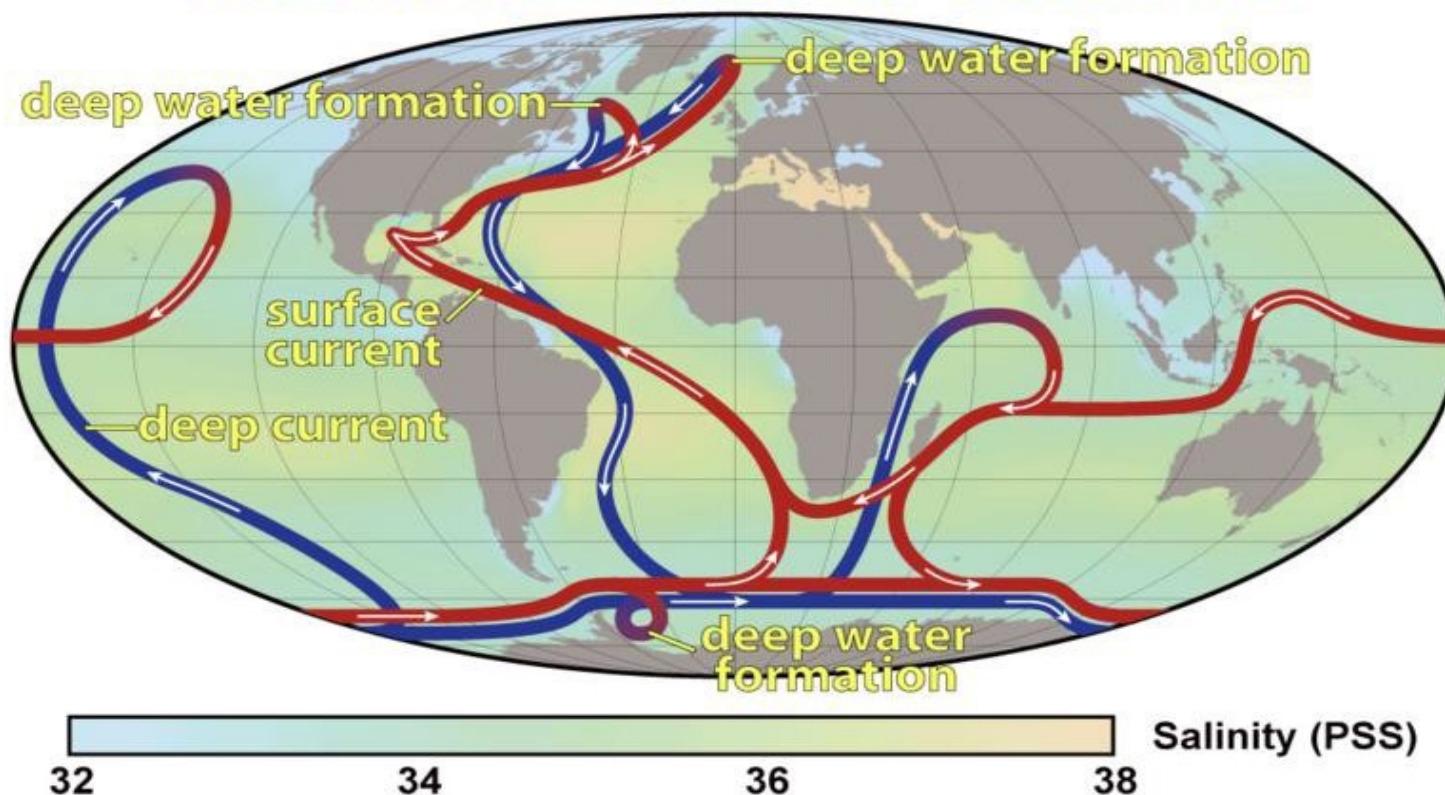
However, in reality Rotation of the earth and the Shape of the Ocean Basin Will also influence the Thermohaline circulation in the ocean.

In Real, Ocean water sinks in the North Atlantic and upwells (rises) in the Pacific and Indian Oceans setting up a **Global Scale Circulation** popularly known as **Conveyor Belt** or Thermohaline Circulation.

The velocity associated with this circulation is typically a few cm/s and the time scale associated with it is a few centuries.

A Simplified view of the Ocean Conveyor Belt (Thermohaline Circulation).

Thermohaline Circulation



Coriolis Force

- The Coriolis force is an apparent force caused by the earth's rotation. The Coriolis force is responsible for deflecting winds/ocean current towards the right in the northern hemisphere and towards the left in the southern hemisphere.
- The Coriolis force acts perpendicular to the pressure gradient force. The pressure gradient force is perpendicular to an isobar. The higher the pressure gradient force, the more is the velocity of the wind and the larger is the deflection in the direction of wind.
- It vanishes at equator, increases with latitude ϕ , maximal at the poles.
- This increases in proportion to the speed V of the object or air parcel.

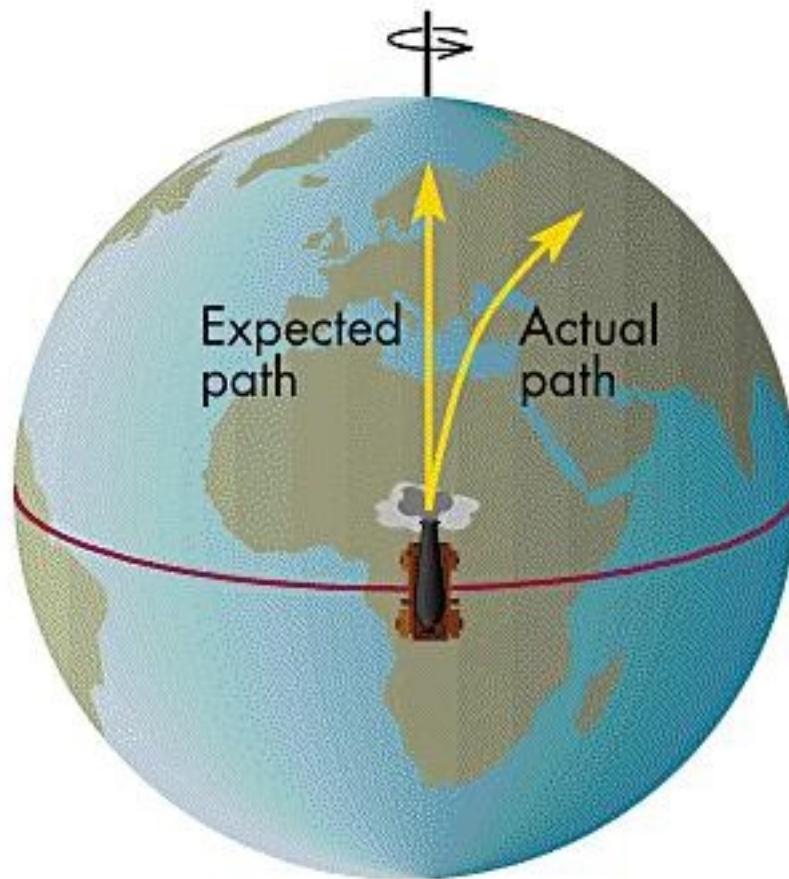
What is Coriolis Effect?

- The Coriolis Effect is an apparent effect produced by a rotating frame of reference.
- The effect occurs when an object moving along a straight path is viewed from a non-fixed frame of reference.
- The moving frame of reference is the Earth which rotates at a fixed speed. Therefore, when an object moving in a straight path is viewed from Earth, it appears to lose its course because of Earth's rotation.
- The Coriolis Effect, therefore, makes things (like planes or currents of water and air) traveling long distances around Earth appear to move at a curve as opposed to a straight line.

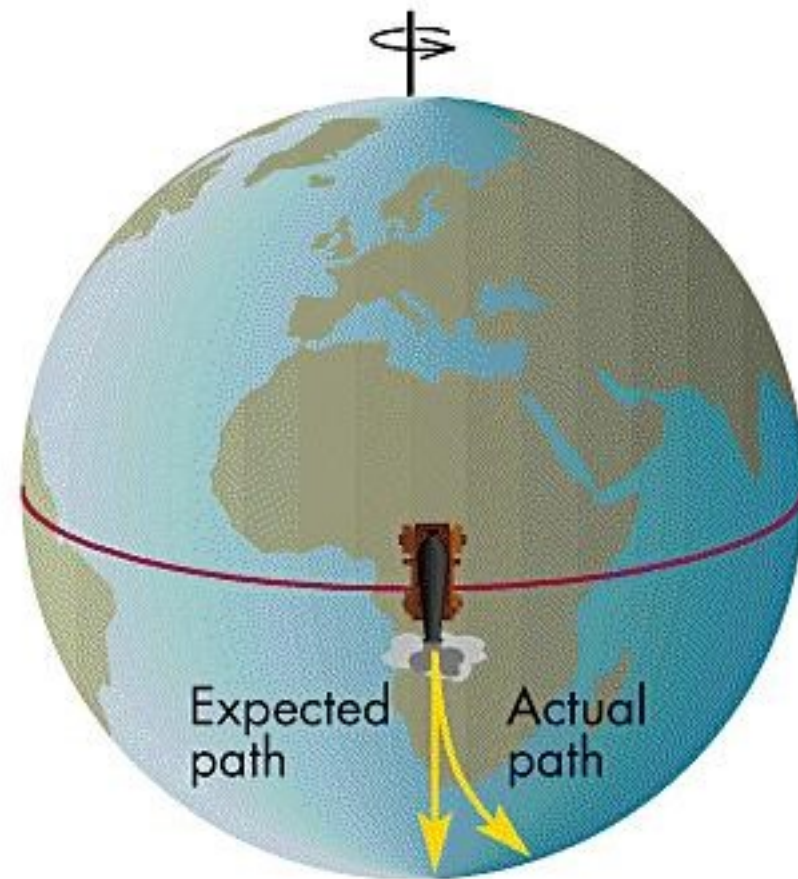
Coriolis Effect

- Coriolis Effect – the tendency of objects moving over the earth (air, water, planes, projectiles) to be deflected (curve away) from a straight line path. The French scientist, Gaspard Coriolis, first explained the deflection of objects moving over the surface due to Earth's rotation
- The deflection is to the right in the Northern Hemisphere and to the left in the Southern Hemisphere.
- This deflection occurs because Earth's surface is rotating with respect to the objects.

Coriolis Effect

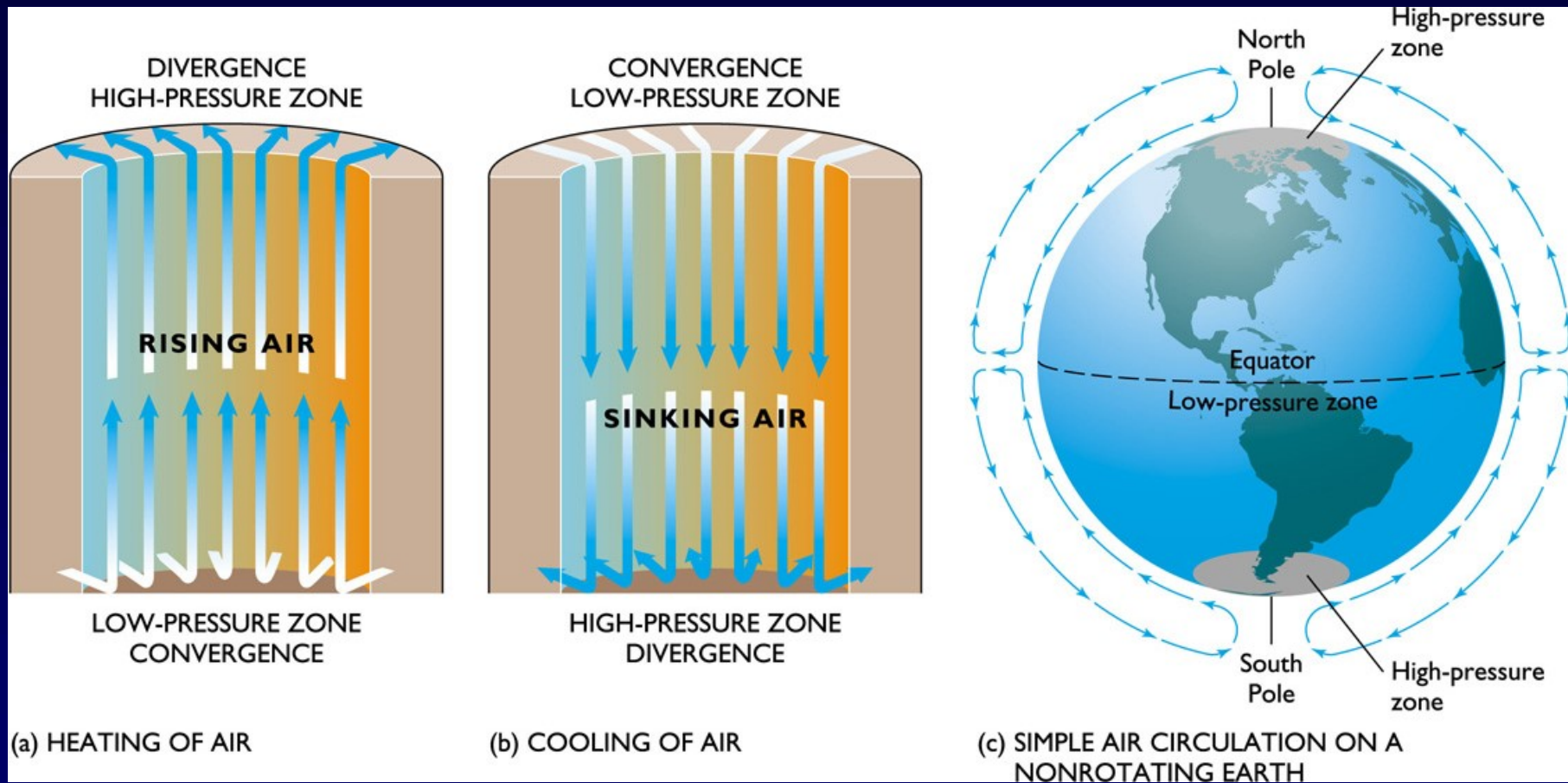


A Projectile fired northward



B Projectile fired southward

Air Circulation on a Non-Rotating Earth



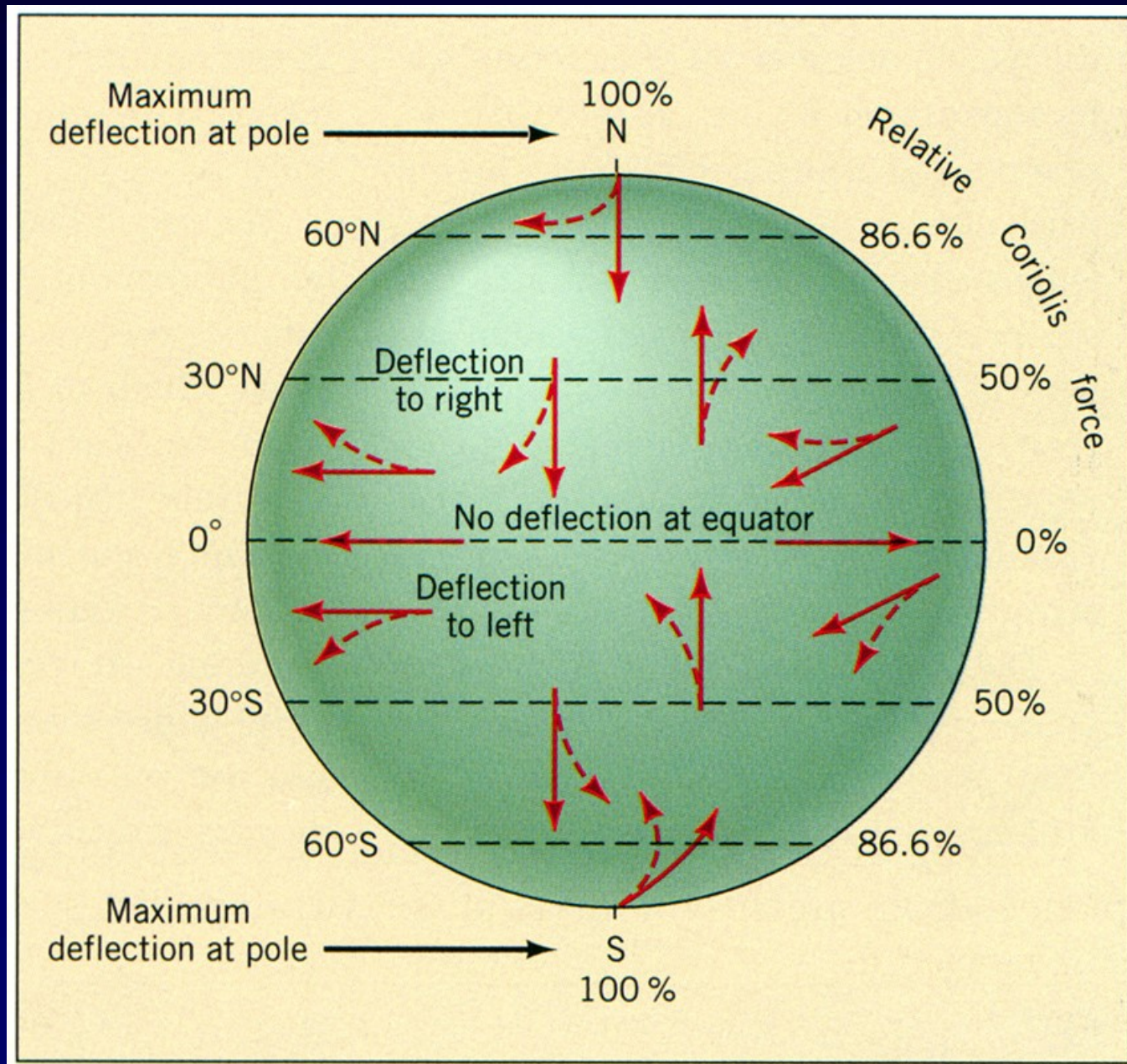
Characteristics of Coriolis Effect

- Coriolis force is a fictitious force resulting from the rotational movement of the earth.
- Coriolis effect is effective on objects that are in motion such as wind, ocean currents, aircraft, ballistic and flying birds.
- Coriolis effect only affects the wind/ocean current direction and not the speed as it deflects the wind/ocean current direction from the expected path.
- The magnitude of Coriolis effect determined by wind/ocean current speed. The higher the speed, the greater the deflection.
- Coriolis effect is maximum at the poles and zero at the equator.
- Coriolis force always acts in a direction that is perpendicular to the moving object's axis.

Forces affecting the wind / ocean water current

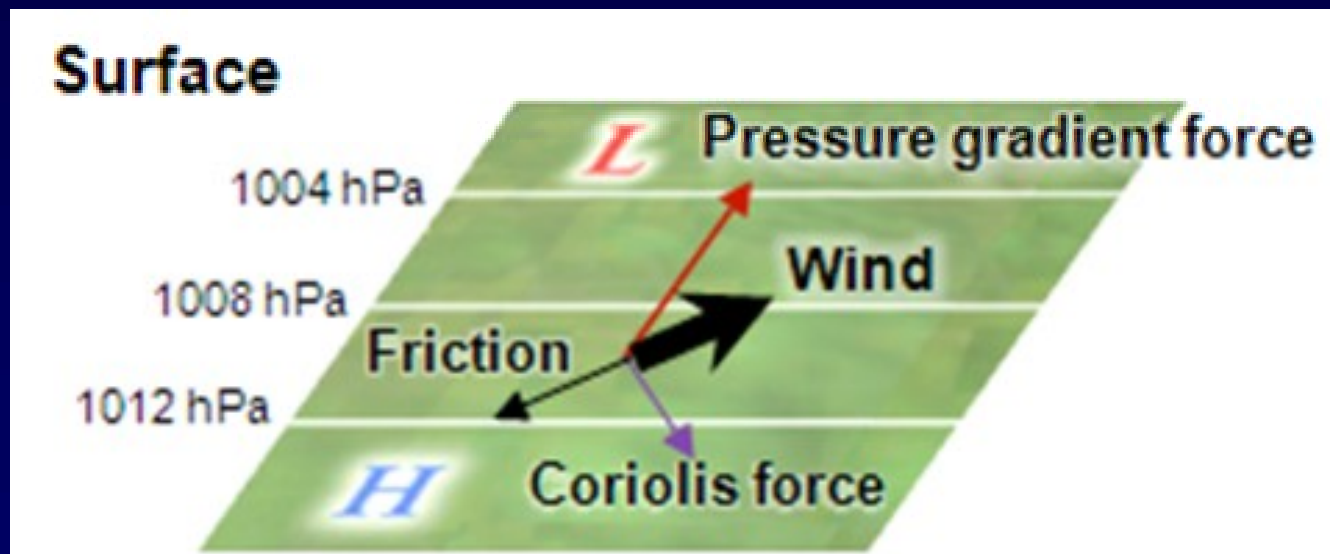
- Pressure-gradient force (PGF), i.e. difference in pressure per unit of distance. The pressure gradient force is the force directed from higher to lower pressure at right angles to the isobars.
- Coriolis force (CF): It is an apparent force that is due to the rotation of the Earth. All free moving objects (winds, currents, rockets, etc.) are deflected to the right of their path in the northern hemisphere and to the left in the southern hemisphere.

Coriolis Force and Coriolis Effect



Forces affecting the wind / ocean water current

- Friction force (FF): Below an altitude of 600 m, friction with the Earth's surface is an important factor in wind. Friction lowers the wind speed and reduces the Coriolis effect.



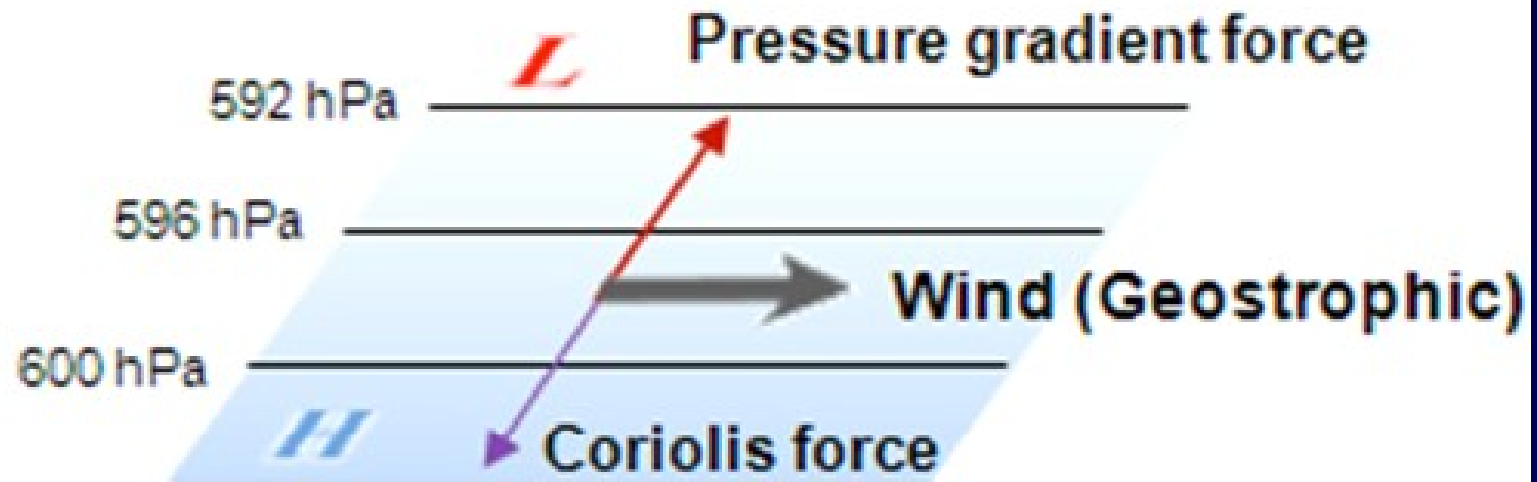
- Gravity/buoyancy force – influences vertical wind / ocean current only.

Geostrophic Wind

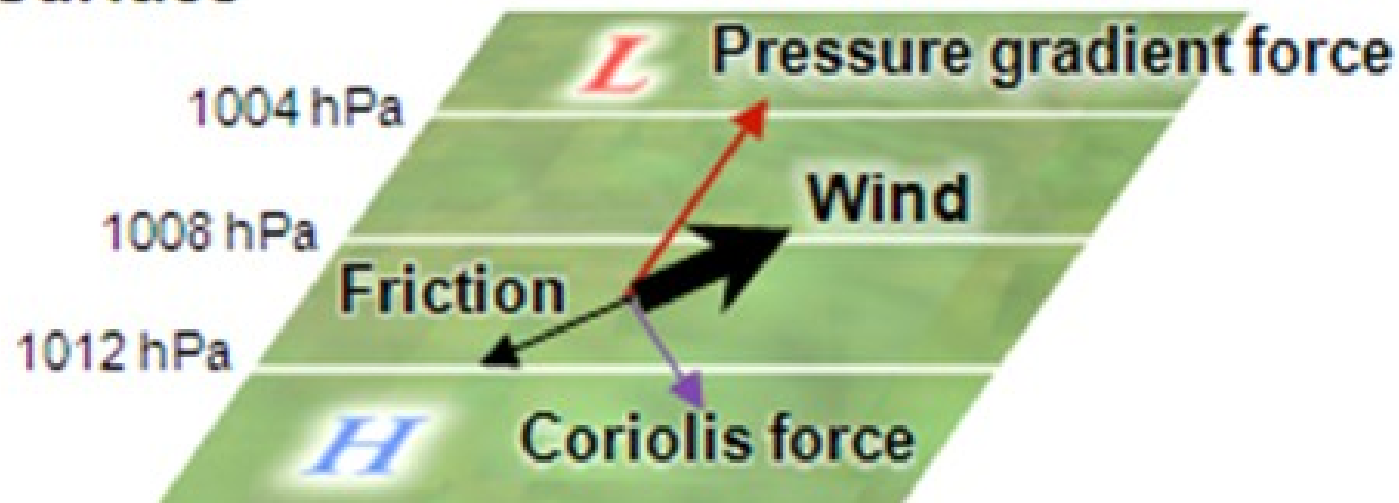
- Wind at elevations above 600 m are above the friction caused by the surface of the Earth.
- Wind direction is a compromise between the PGF (across isobars) and the Coriolis force (deflection to the right) and winds flow parallel to isobars.
- Geostrophic wind is wind at high altitude (above the friction of the Earth's surface) that flows in a path parallel to isobars.
- The geostrophic wind describes the dominant balance between the pressure gradient force and the Coriolis force.
- The closer the isobars are together, the stronger the magnitude of the geostrophic wind.

Geostrophic Wind

Free Atmosphere (no friction)

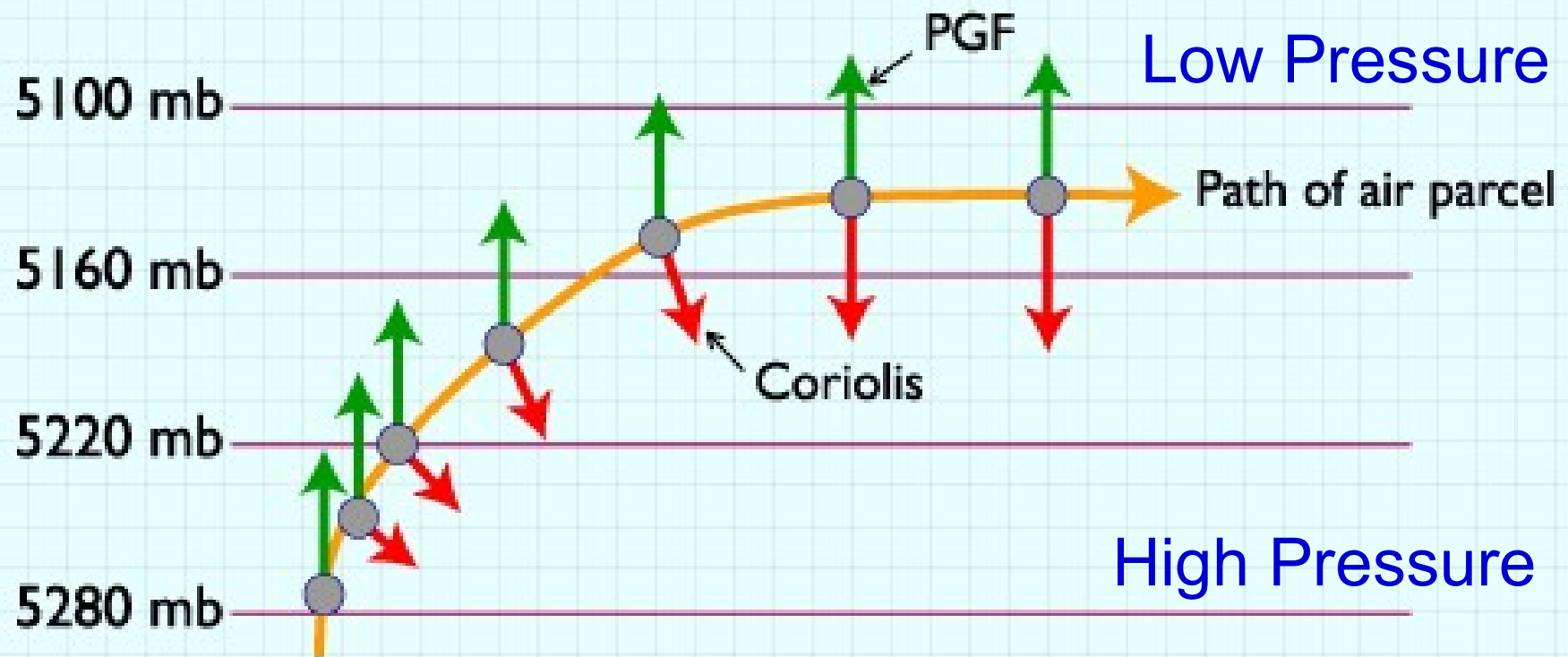


Surface



Geostrophic Wind

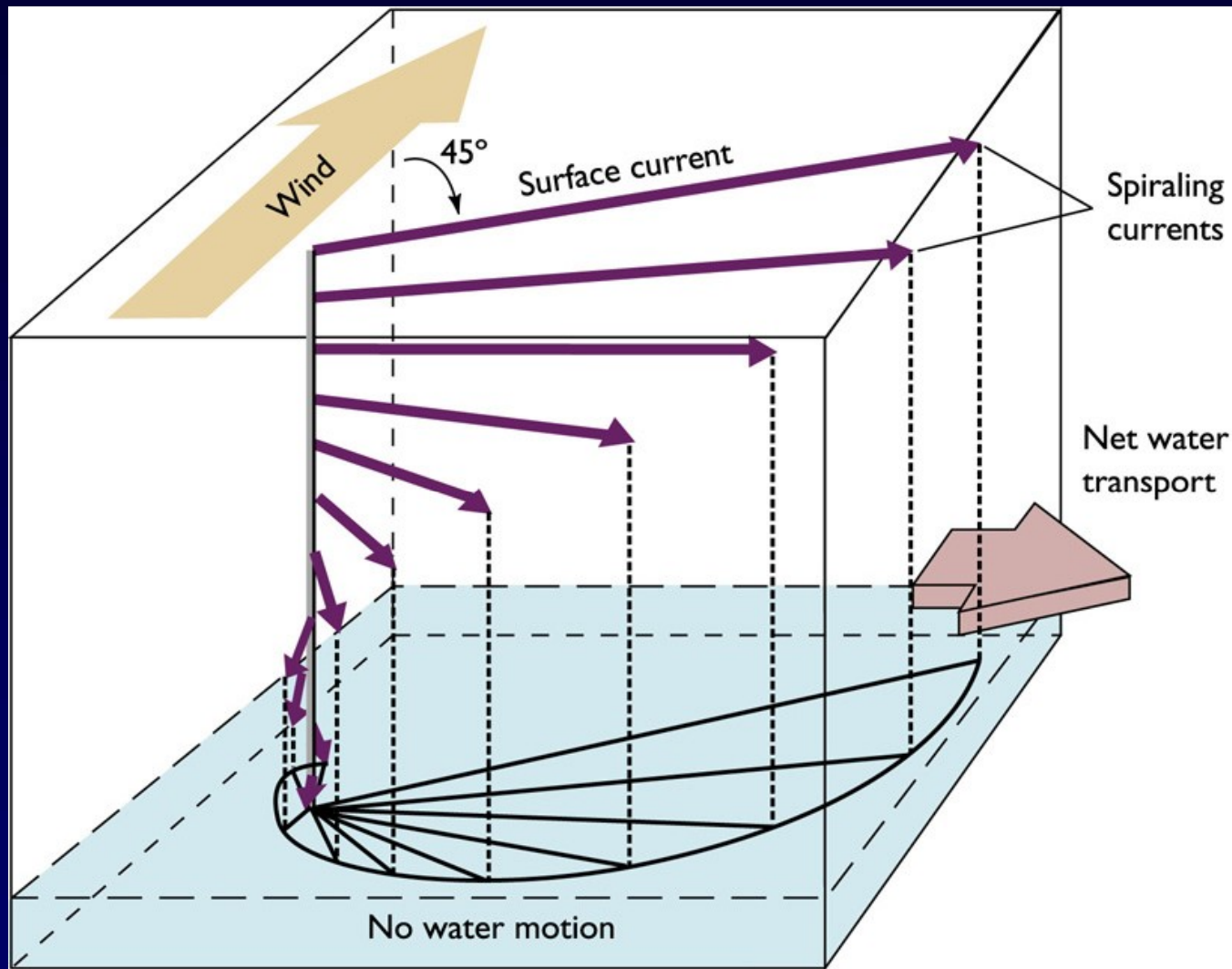
Geostrophic Wind



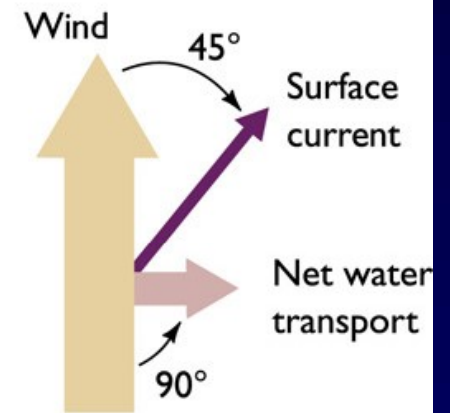
Ekman Transport

- Ekman transport (first investigated in 1902 by Vagn Walfrid Ekman) occurs when ocean surface waters are influenced by the friction force acting on them via the wind. As the wind blows it casts a friction force on the ocean surface that drags the upper 10-100m of the water column with it.
- However, due to the influence of the Coriolis effect, the ocean water moves at a 90° angle from the direction of the surface wind.
- The direction of transport is dependent on the hemisphere: in the northern hemisphere, transport occurs at 90° clockwise from wind direction, while in the southern hemisphere it occurs at 90° anticlockwise.
- Ekman transport has significant impacts on the biogeochemical properties of the world's oceans. This is because it leads to upwelling (Ekman suction) and downwelling (Ekman pumping) in order to obey mass conservation laws. Mass conservation, in reference to Ekman transfer, requires that any water displaced within an area must be replenished. This can be done by either Ekman suction or Ekman pumping depending on wind patterns.

Ekman Transport



(a) EKMAN SPIRAL IN THE NORTHERN HEMISPHERE

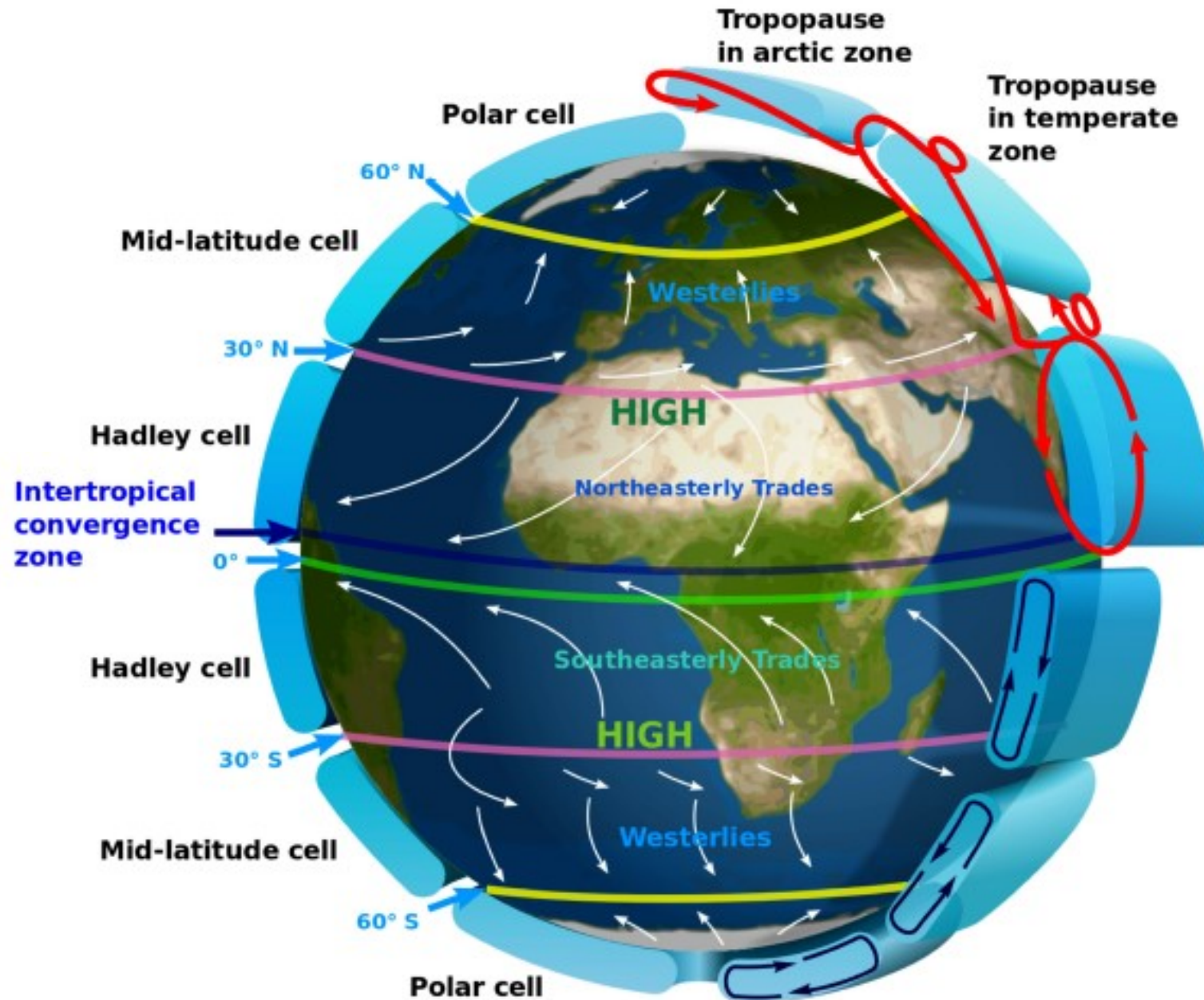


(b) MAPVIEW

Trade Wind

The **trade winds** or **easterlies** are air currents closer to Earth's surface and are the permanent east-to-west prevailing winds that flow in the Earth's equatorial region. It primarily blows steadily towards the equator from the north-east in the northern hemisphere or the south-east in the southern hemisphere, especially at sea. Two belts of trade winds encircle the earth, blowing from the tropical high-pressure belts to the low-pressure zone at the equator.

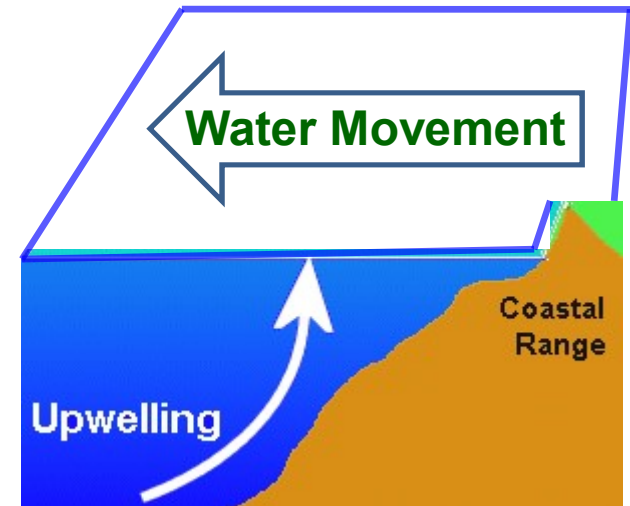
Trade Wind



Upwelling

What is upwelling?

Upwelling is a process in which deep, cold water rises toward the surface.

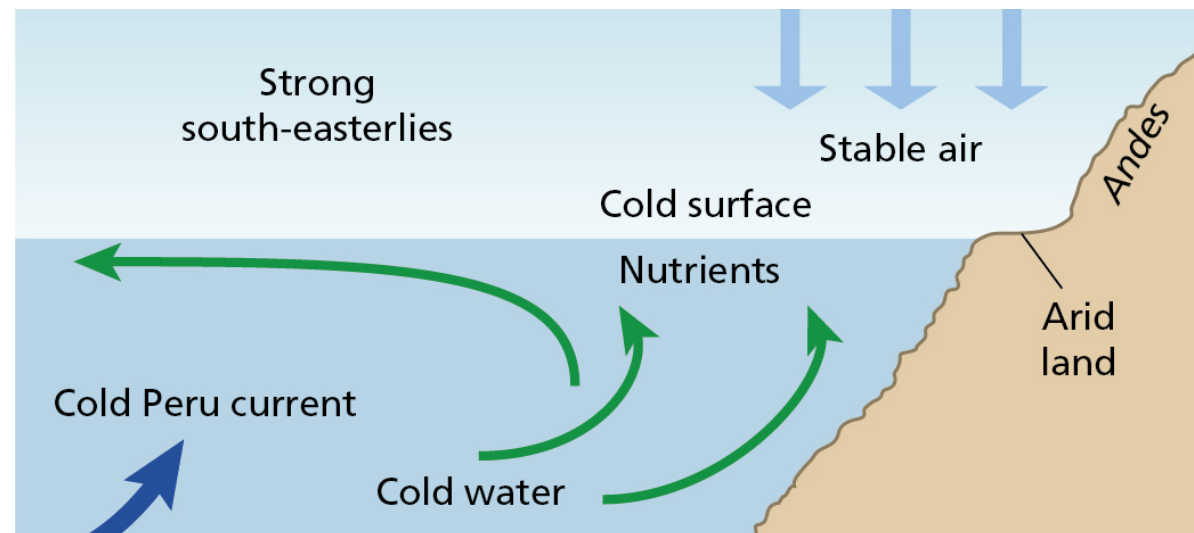
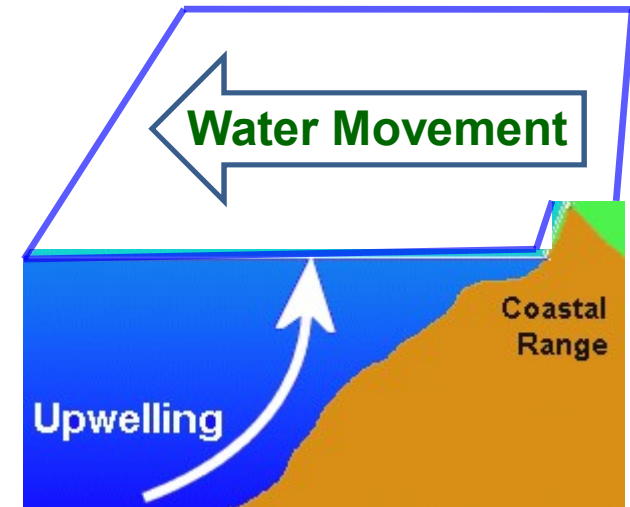


What is upwelling?

Upwelling is a process in which deep, cold water rises toward the surface.

How it occurs?

When winds blowing across the ocean surface push water away from the coast, Water then rises up from beneath the surface to replace the water that was pushed away. This process is known as “upwelling.”



What is upwelling?

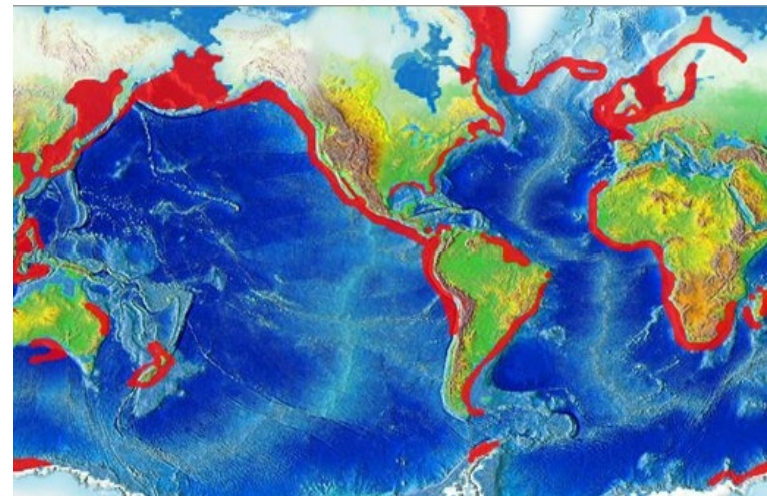
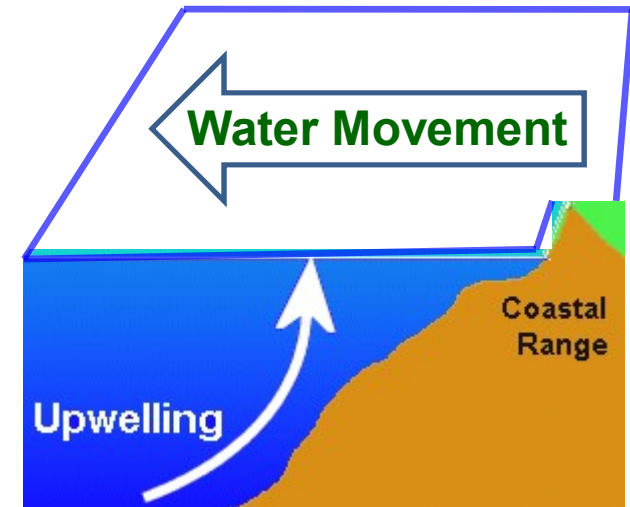
Upwelling is a process in which deep, cold water rises toward the surface.

How it occurs?

When winds blowing across the ocean surface push water away from the coast, Water then rises up from beneath the surface to replace the water that was pushed away. This process is known as “upwelling.”

Where it occurs?

Upwelling occurs in a number of places in the global ocean mostly along the coastal boundary



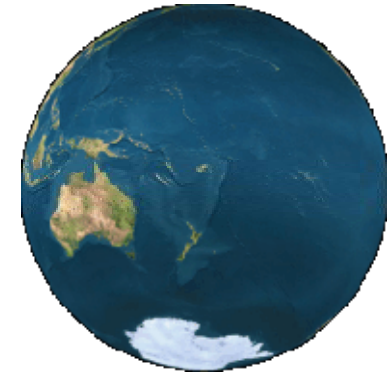
Types of Upwelling

There are three main types of upwelling

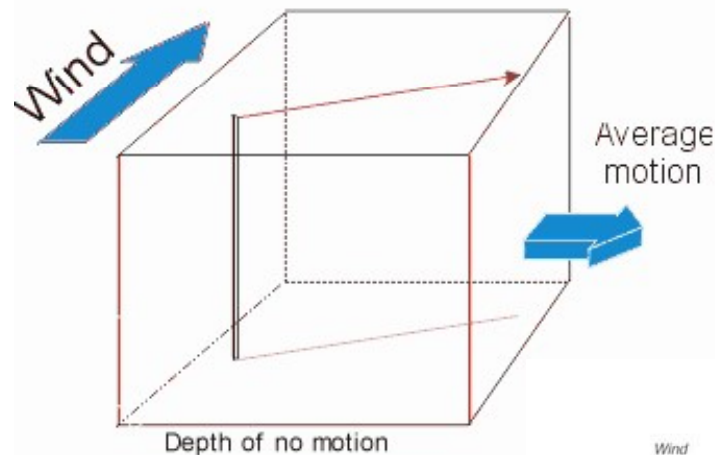
- 1. Coastal Upwelling**
- 2. Equatorial Upwelling**
- 3. Open Ocean Upwelling**

Let us Examine What Drives each of these Upwelling

Mechanisms that Create Ocean Upwelling

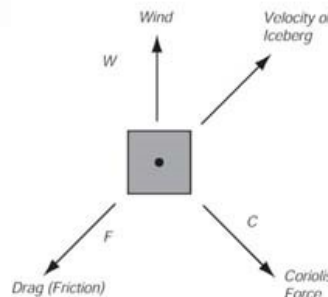
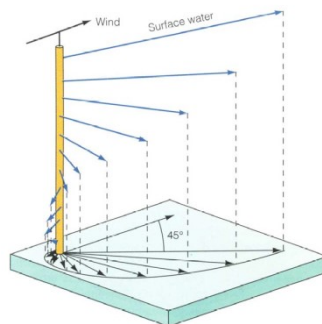


In order to understand the different types of upwelling, the Coriolis Effect and Ekman Transport need to be understood.



In the Ekman layer the frictional force is balanced by the Coriolis force.

The mean mass transport in the Ekman Layer is 90° to the right of the wind direction in the northern hemisphere



Ekman layer depth

$$V_0 = \frac{0.0127}{\sqrt{\sin |\varphi|}} U_{10},$$

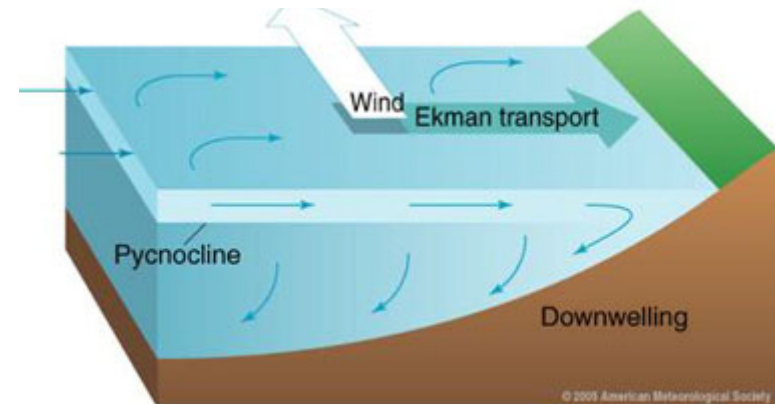
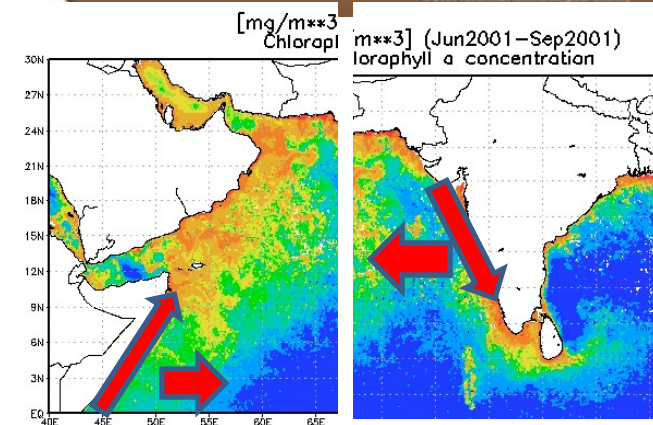
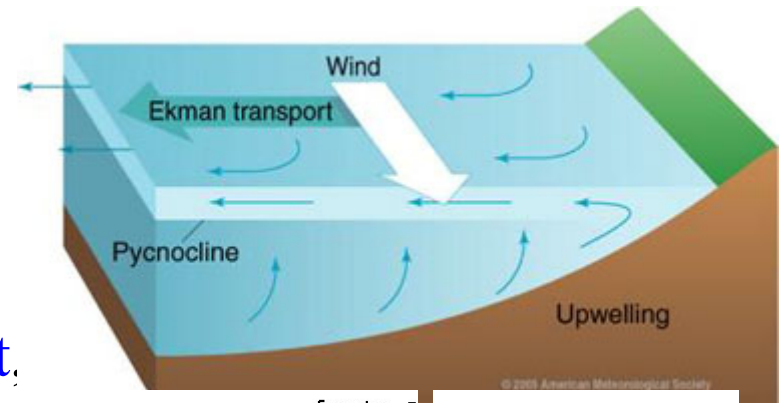
$$D_E = \frac{7.6}{\sqrt{\sin |\varphi|}} U_{10}$$

1.Coastal Upwelling

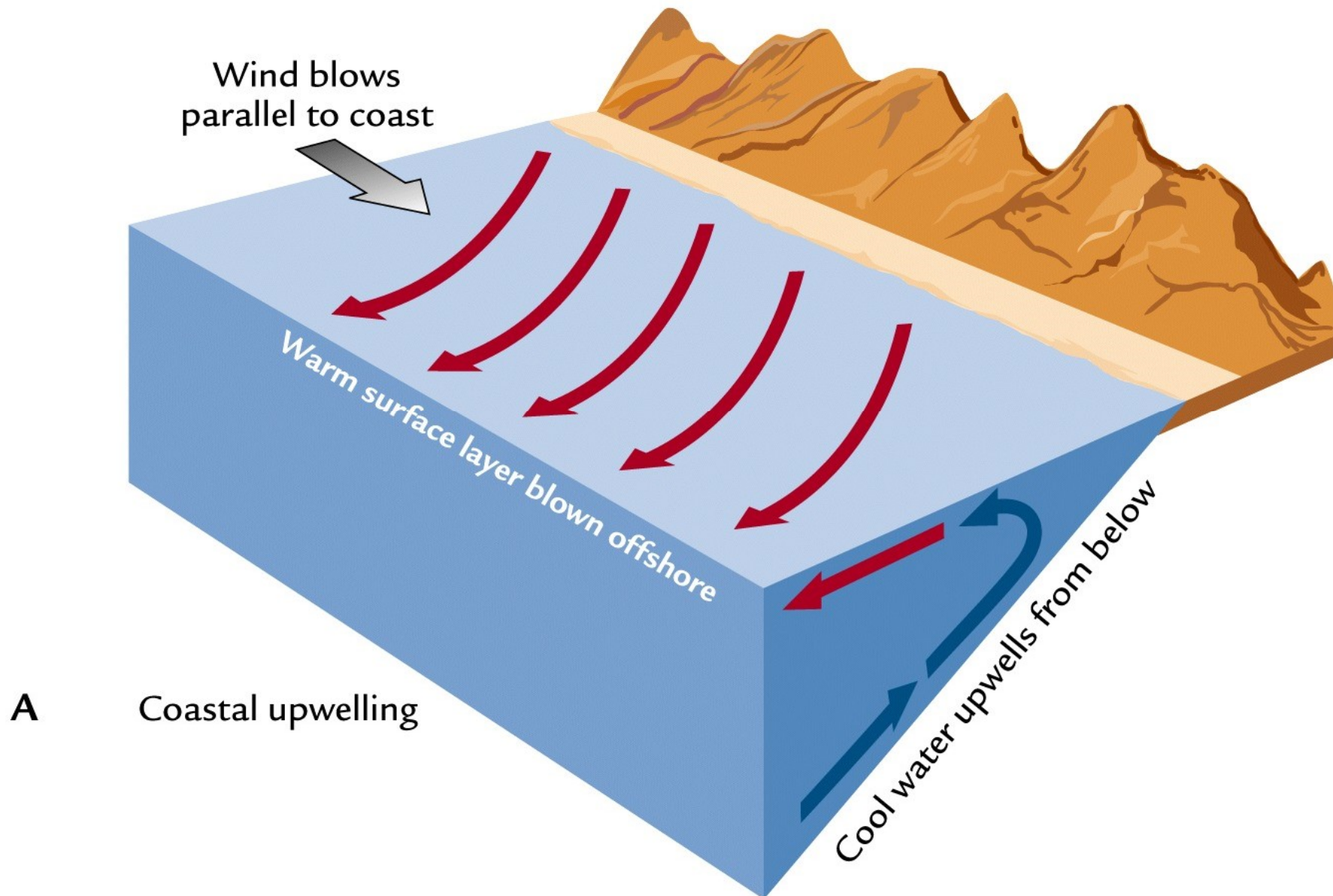
Now Let US Understand the Coastal Upwelling?

In the **Northern-Hemisphere** When a **Equator ward** (**Pole ward**) wind blows along the coast **with the coast to its left**, then the Ekman Transport is to the right of the wind or **away** from the coast. As the coastal area has a deficit of water (mass), upwelling occur (mass conservation). This is a case similar to the coastal **upwelling** along the southwest coast of India (**Somalia**) during **summer monsoon**

The reverse process, called “**downwelling**,” also occurs when wind causes surface water to **build up along a coastline** and the surface water eventually sinks toward the bottom.

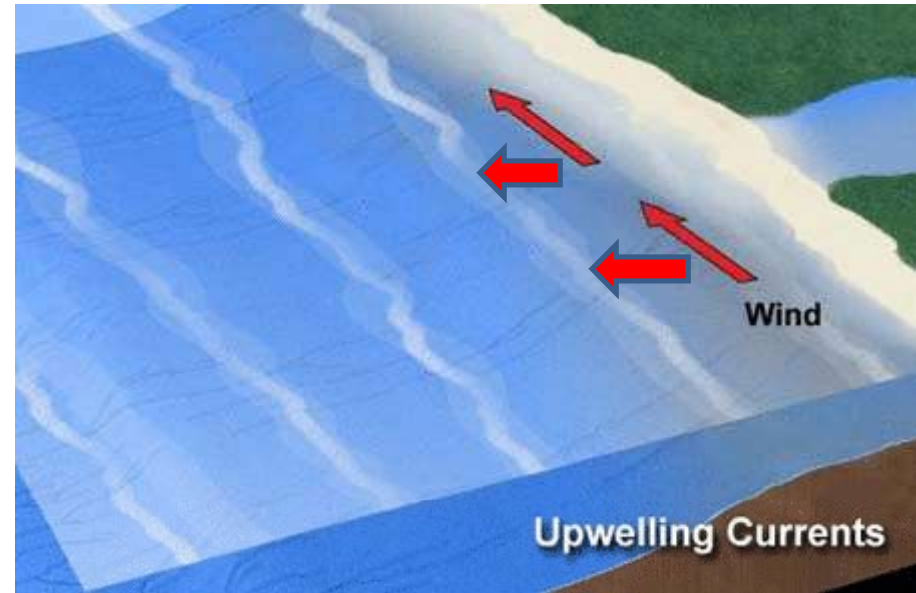


Coastal Upwelling



Coastal Upwelling In The Southern-Hemisphere

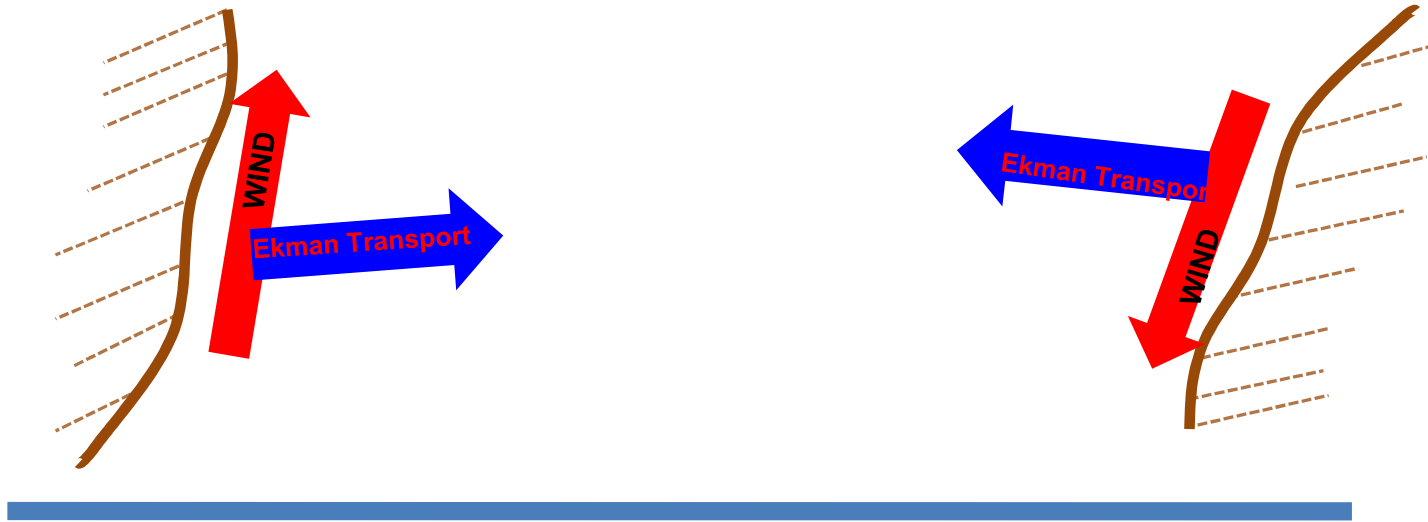
In the **Southern-Hemisphere** When a **Equator ward** (**Pole ward**) wind blows along the coast **with the coast to its right** then the Ekman Transport is to the **left** of the wind or **away** from the coast. As the coastal area has a deficit of water (mass), upwelling occur (mass conservation). This is a case similar to the coastal **upwelling** along the Peru coast of South America.



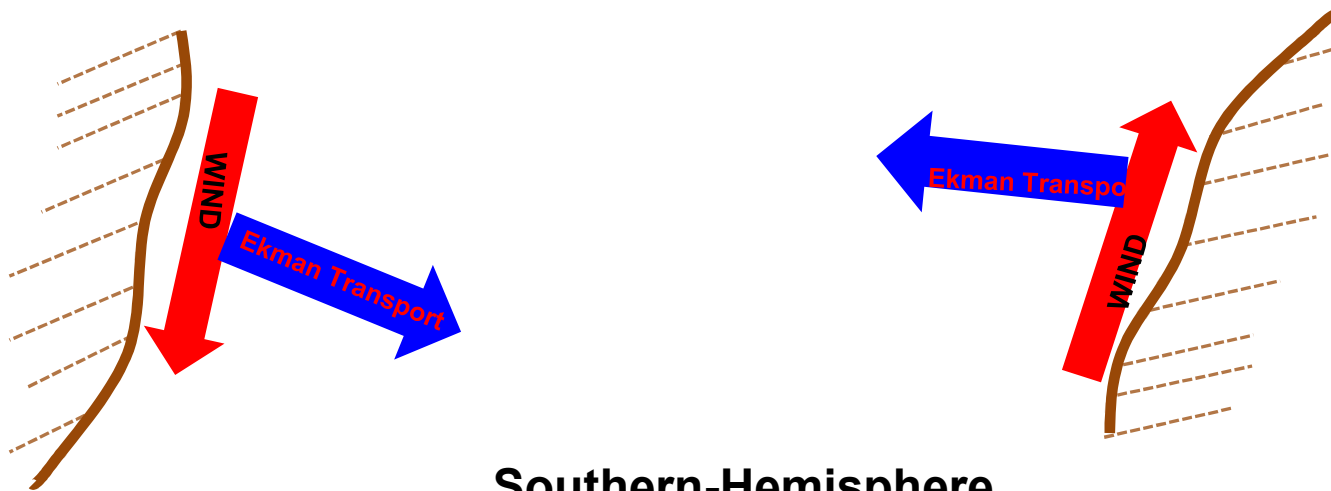
http://en.wikipedia.org/wiki/File:Upwelling_animated.gif#filelinks



Northern-Hemisphere



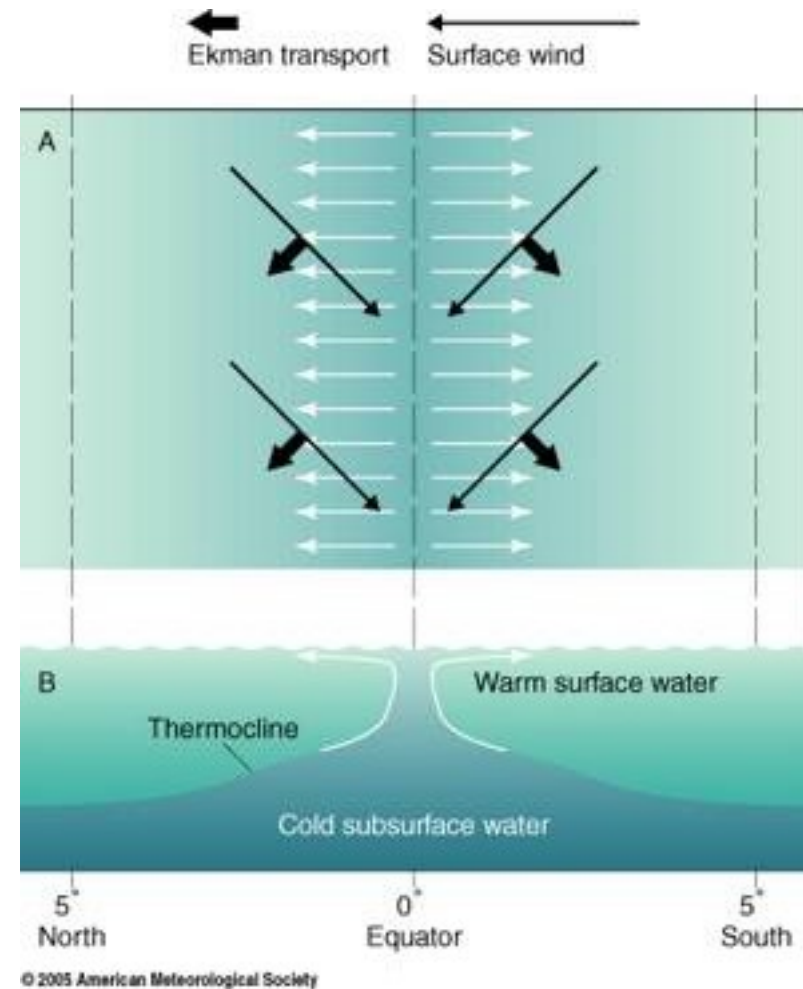
Southern-Hemisphere



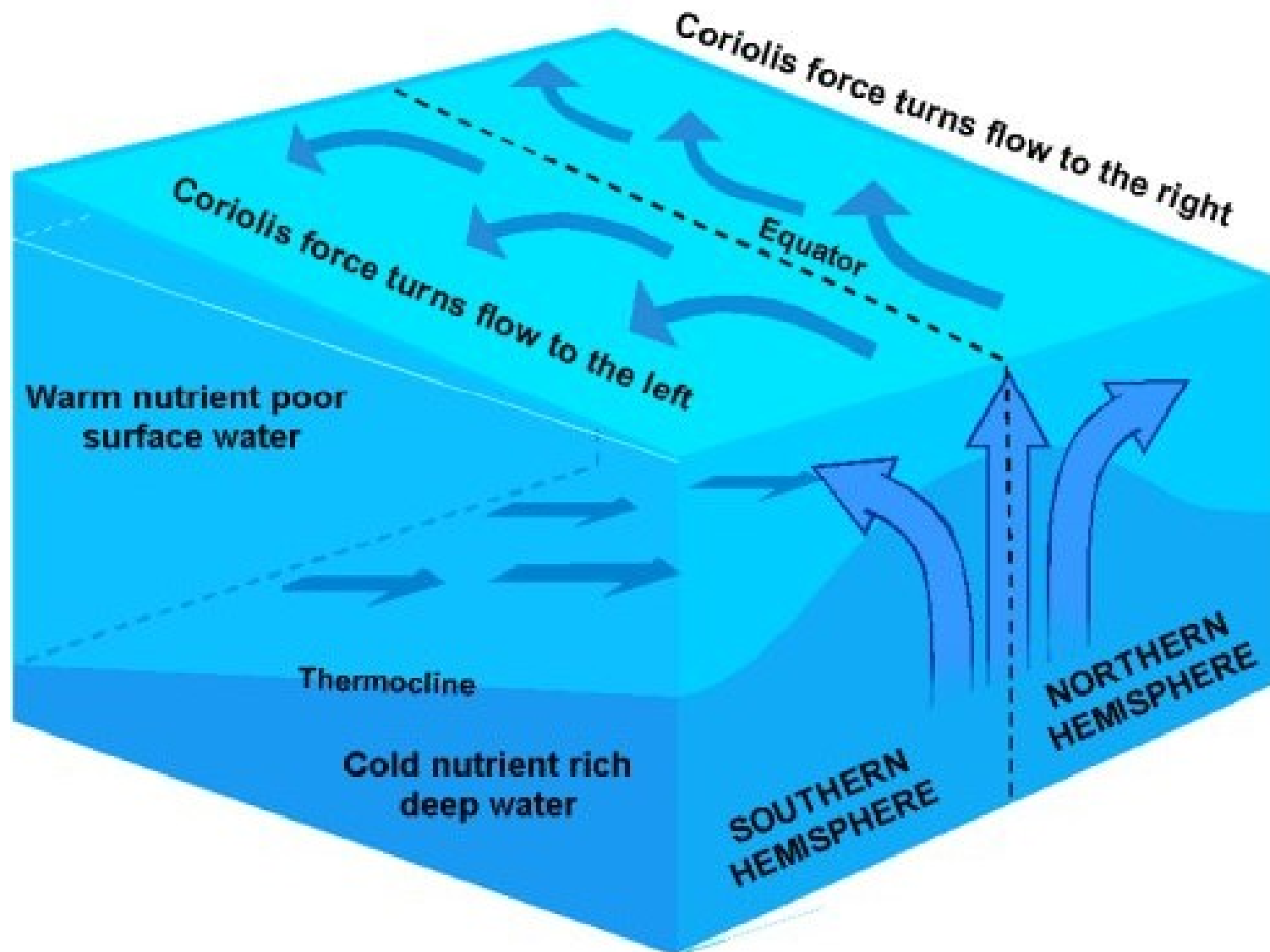
2. Equatorial Upwelling

Equatorial Upwelling?

Equatorial upwelling is caused by the winds known as the **trade winds**. The trade winds blow from **east to west** in the vicinity of the equator. On the northern side of the equator **Ekman Transport** is to the **right** (northward), and on the southern side it is to the **left** (southward). With water flowing directly away from the equator, both northward and southward, the equator itself has a **deficit of water**. Hence, water from below upwells to fill in the gap. Equatorial upwelling is most prominent in the **Pacific Ocean**.



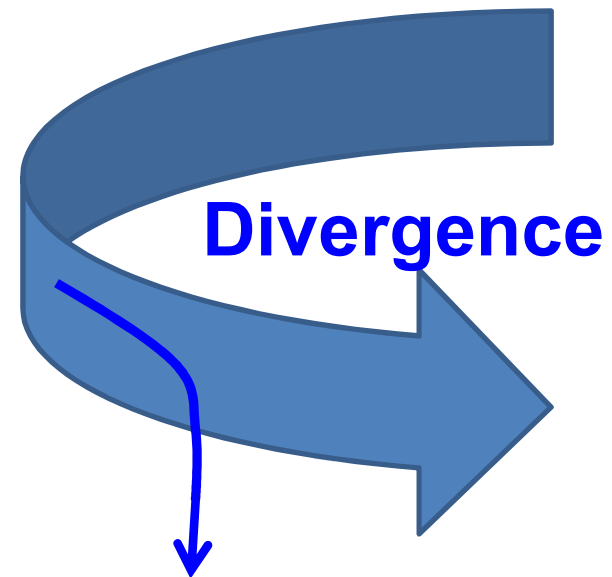
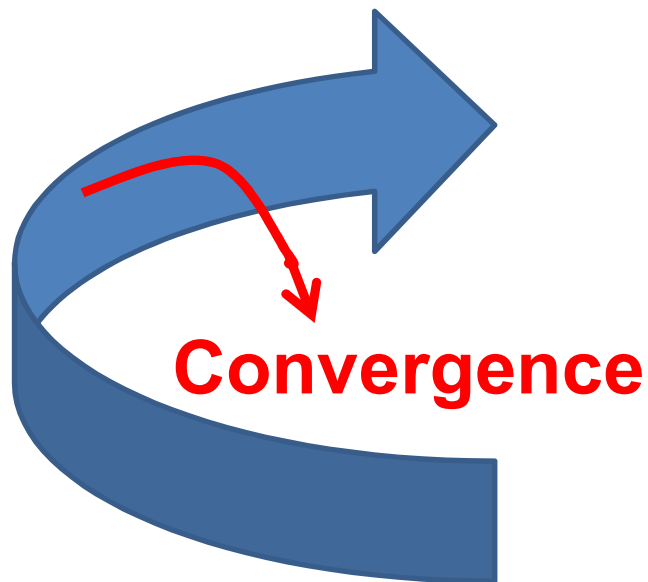
Equatorial Upwelling



3. Open Ocean Upwelling

Open Ocean Upwelling

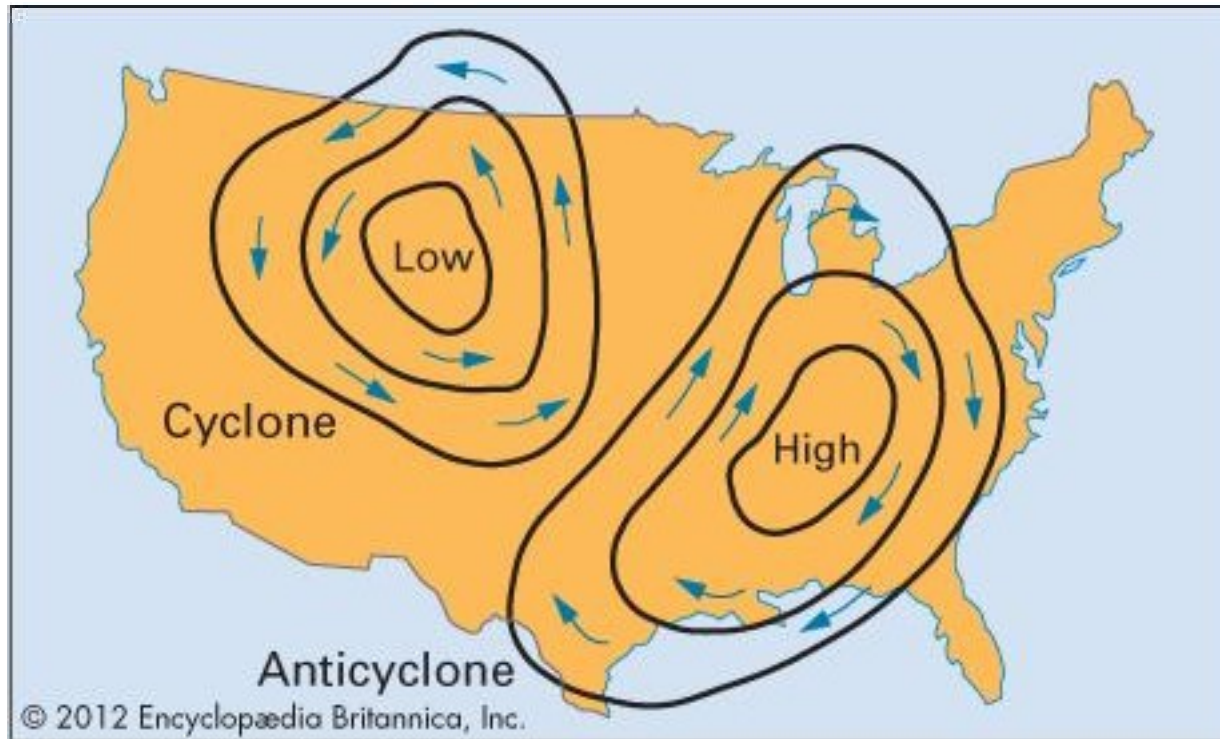
- Storms or cyclones that occur seasonally in the open ocean also contribute to areas of upwelling. As the storm or cyclone winds move over the ocean, temporary upwelling occurs due to the storm's movement over the surface water.

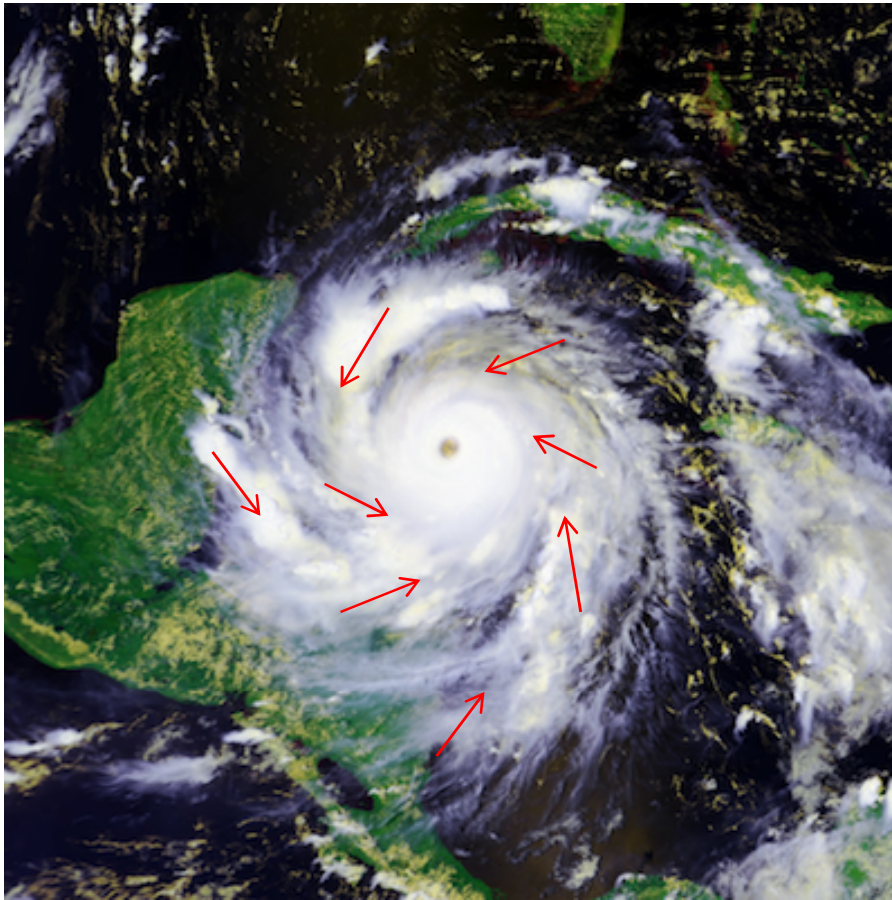


Cyclonic and Anticyclonic Winds

Cyclones are low pressure systems - winds blow inward and counterclockwise around a low in the N. hemisphere.

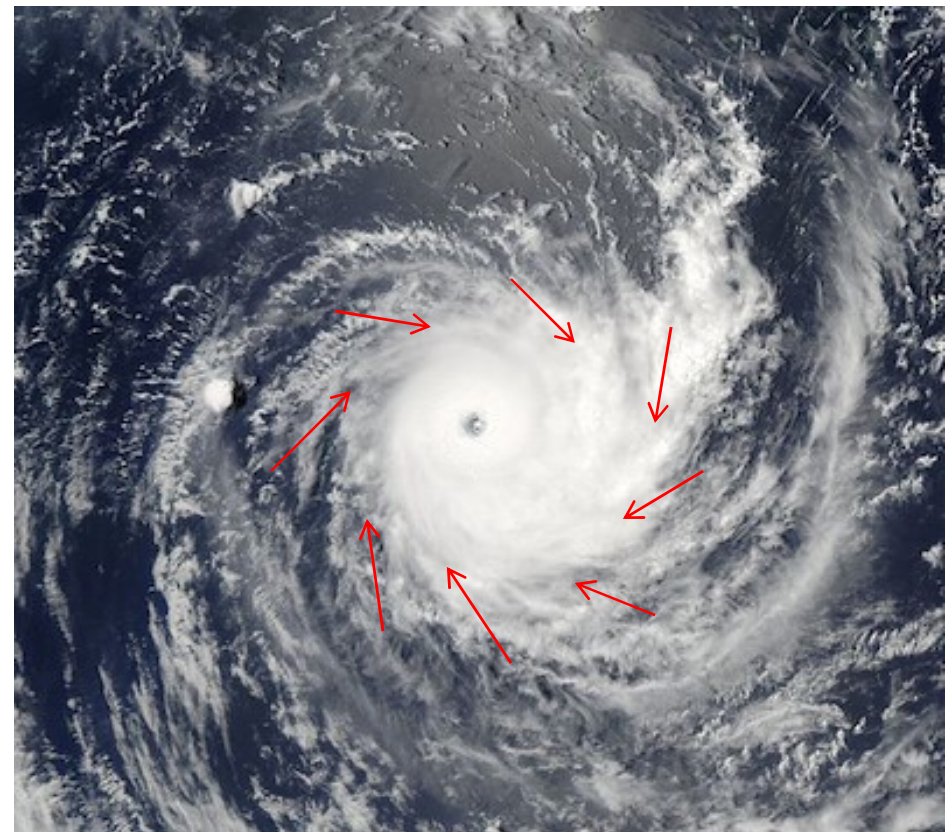
Anticyclones are high pressure systems - winds blow outward and clockwise out of the high in the N. hemisphere.





Satellite photo of a cyclone (low pressure system) in the northern hemisphere with counterclockwise circulation.

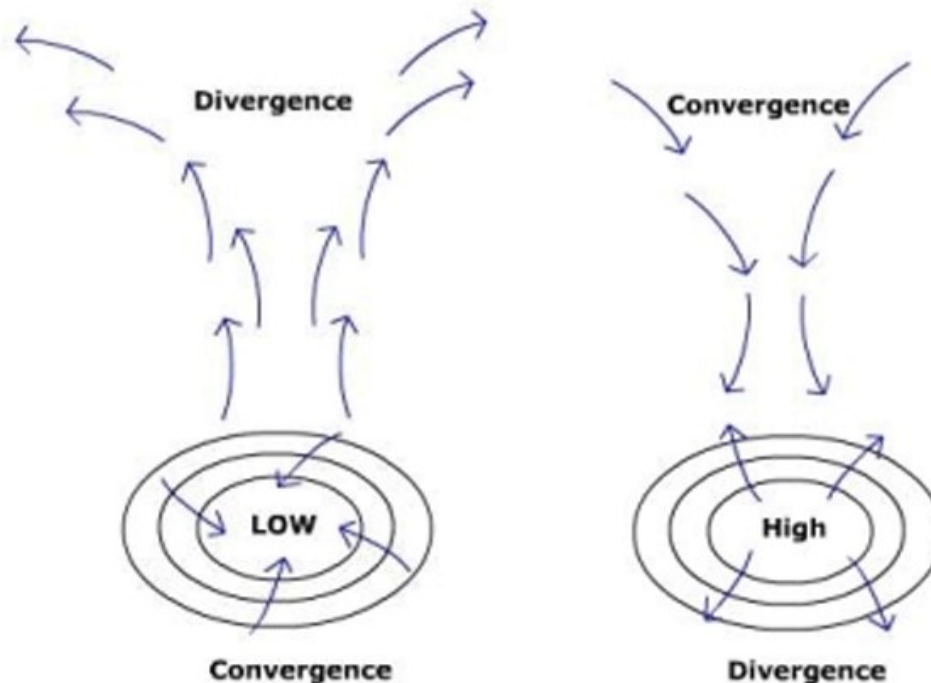
Satellite photo of a cyclone (low pressure system) in the southern hemisphere with clockwise circulation.



Cyclones and Anticyclones

At the surface as winds blow inward toward the center of a low, the air must go somewhere. The *converging* air at the surface slowly rises. Aloft, the upflow will *diverge* (spread apart) to compensate for the *converging* surface air.

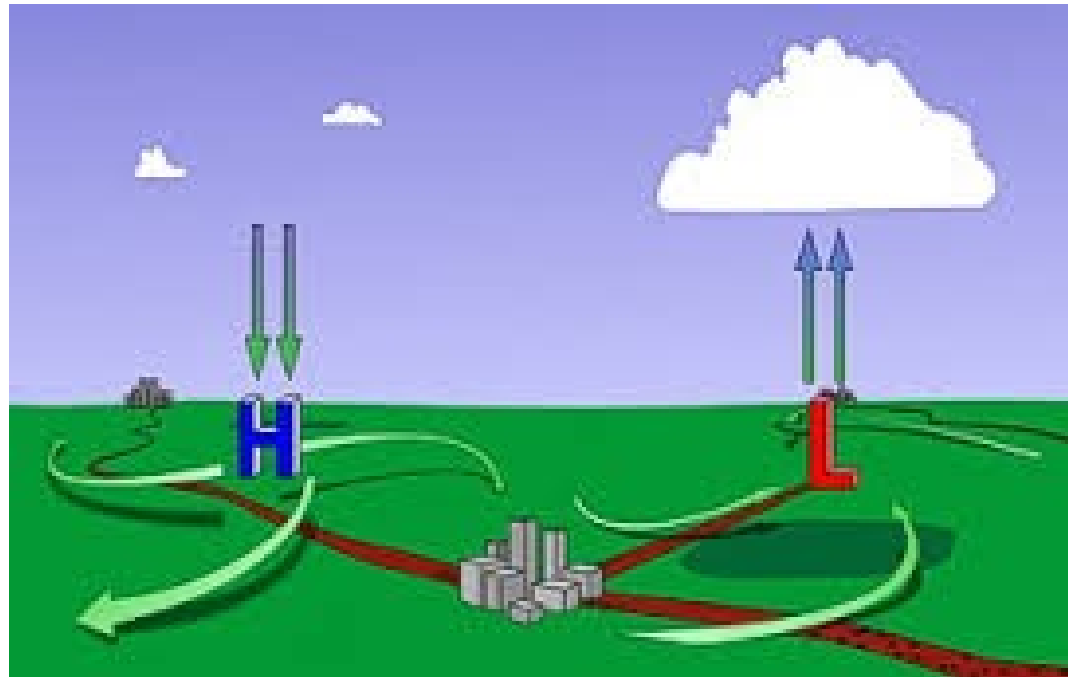
Surface pressure will change if the upper-level outflow and the surface inflow are not balanced.



Cyclones and Anticyclones

Descending air associated with a high pressure system is associated with atmospheric stability and clear conditions.

Ascending air associated with a low pressure system is associated with atmospheric instability, cloud development and precipitation.

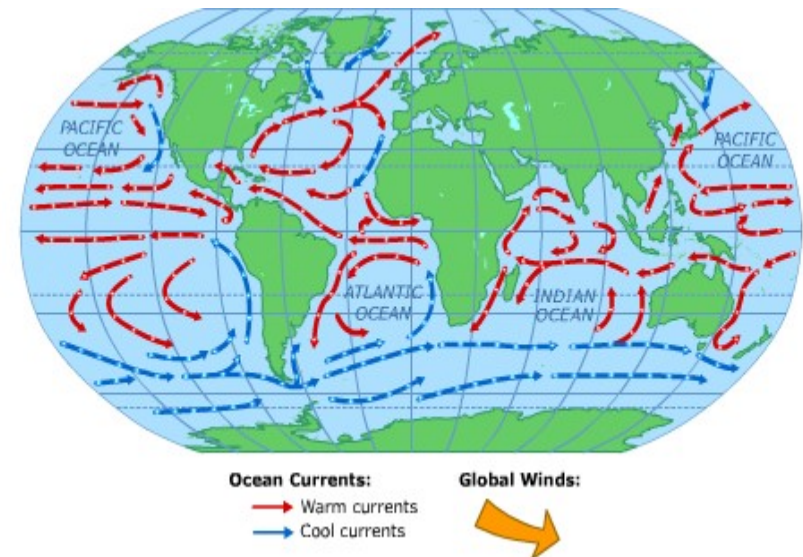
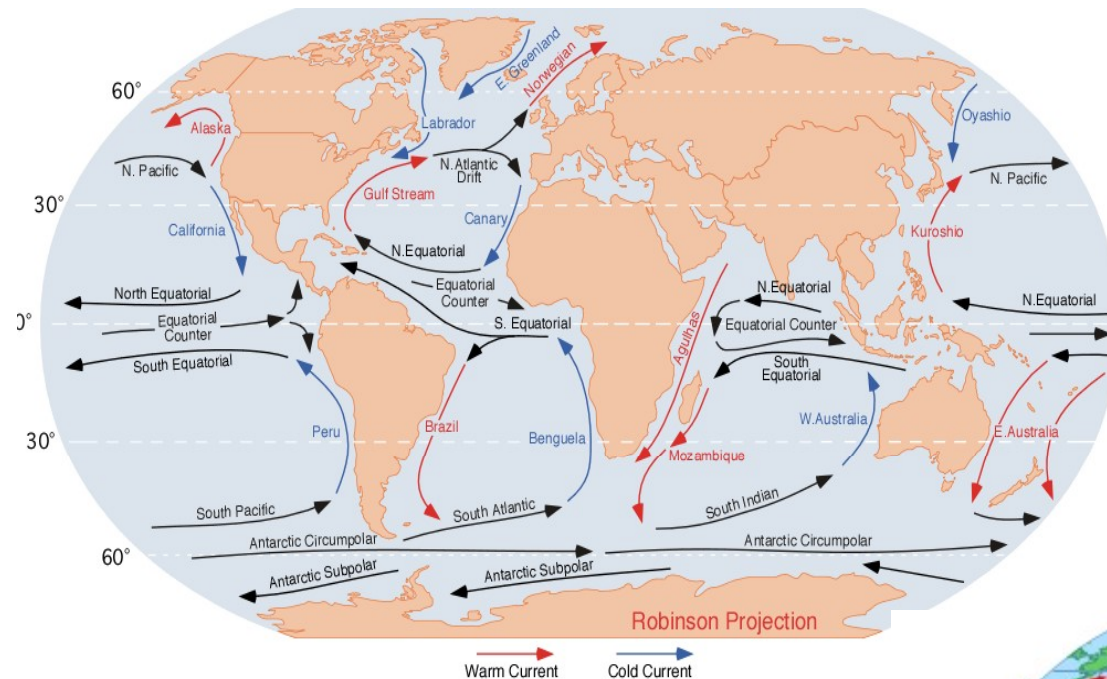


Major Zones of Upwelling

Major zones of upwelling occur along the western coasts that receive cold currents from the poles. Cold currents and their locations are highlighted in the following list, showing examples of upwelling zones.:

1. Benguela Current - located off the southwest coast of Africa.
2. Canary Current- located off the northwest coast of Africa.
3. California Current - located off the northwest coast of the U.S.
4. Humboldt Current - located off the coast of Peru and Chili.
5. Somali Current - located off the coast of Somalia and Oman.

Locate the regions of intense Upwelling in the Global Ocean



Is upwelling good or bad?

- The Water that rises to the surface as a result of upwelling is typically colder and is rich in nutrients.
- Upwelling is a positive boost to the marine ecosystem by providing nutrients to enhance plankton growth, which is the basis of the ocean food chain.
- Therefore, good fishing grounds typically are found where upwelling is common.
- However, in areas where upwelling occurs, the surface water that is moving away from the shore can take surface organisms with it, which may reduce the biological productivity in that area.

What causes upwelling?

- Upwelling occurs due to wind, the Coriolis effect, and the presence of deep currents.
- As the wind blows on the surface water and displaces it outward, deep current water moves upward to fill in the space.
- The rotation of the Earth causes water to move off shore along the western continental coasts, which is a 90-degree angle from the wind direction.



Thank You

Pillow lava at CIR observed during manned submersible dive